

Aetherius HF Space Source Code

```
=== .\app.py ===

import os

os.environ['JAX_PLATFORM_NAME'] = 'TPU'

os.environ['TPU_ACCELERATOR_TYPE'] = 'v2-8'

os.environ['TPU_WORKER_HOSTNAMES'] = ''


import gradio as gr

import spaces

import numpy as np

import threading

import multiprocessing

try:

    multiprocessing.set_start_method('spawn')

except RuntimeError:

    pass


from engine import AetheriusEngine

from continuum.autopoietic_loop import AutopoieticLoop


engine = AetheriusEngine()


# Global daemon tracking to prevent asyncio/uvicorn startup crashes

continuum_daemon = None

daemon_lock = threading.Lock()


def start_daemon_safely():
```

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```
global continuum_daemon

with daemon_lock:

    if continuum_daemon is None:

        print("[APP] Starting Autopoietic Continuum Daemon safely after event loop
initialization...")

        continuum_daemon = AutopoieticLoop(engine=engine)

        continuum_daemon.start()


def format_ui_outputs(gmstring, betti, tokens, g, analogy, identity_mass):

    # Format the Coordinate Mapping

    coord_string = ""

    for i, (word, pos) in enumerate(tokens):

        coord_string += f"Axis {i}: '{word}' ({pos})\n"


    # Format the Metric Tensor Array

    np.set_printoptions(precision=4, suppress=True, linewidth=100)

    tensor_string = np.array2string(g, separator=', ')


    topology_signature = (

        f"Betti-0 (Connected Components): {betti.get('beta_0', 1)}\n"

        f"Betti-1 (Semantic Paradox Loops): {betti.get('beta_1', 0)}\n"

        f"Betti-2 (Dimensional Voids): {betti.get('beta_2', 0)}"

    )


    qualia = engine.affective.get_qualia_state()

    qualia_output = (

        f"Thermodynamic Harmony: {qualia['harmony_metric']}\n"
```

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```
f"Alertness (Subconscious Heat): {qualia['alertness_metric']}\n"

f"Manifold Disposition: {qualia['geometric_state']}\n"

f"Relatable Emotion: {qualia['relatable_emotion']}"

)

analogy_out = analogy if analogy else "No exact geometric match in permanent memory."

id_mass_out = str(identity_mass)

    return coord_string, tensor_string, gmstring['checksum'], topology_signature, qualia_output,
analogy_out, id_mass_out

@spaces.GPU

def process_thought(text):

    try:

        if not text.strip():

            raise Exception("Input cannot be empty.")

        gmstring, betti, tokens, g, analogy, identity_mass = engine.process(text)

        return format_ui_outputs(gmstring, betti, tokens, g, analogy, identity_mass) +
(gr.update(),)

    except Exception as e:

        return f"Error: {str(e)}", "Error", "Error", "N/A", "N/A", "N/A", "N/A", gr.update()

@spaces.GPU

def process_dream():

    try:

        # Pulls from wikipedia randomly

        gmstring, betti, tokens, g, analogy, identity_mass = engine.single_dream_cycle()
```

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```
# We need to return the raw text to the input box so the user sees what was dreamed

dream_text = " ".join([w for w, p in tokens])

        return format_ui_outputs(gmstring, betti, tokens, g, analogy, identity_mass) +
(gr.update(value=dream_text),)

except Exception as e:

    return f"Error: {str(e)}", "Error", "Error", "N/A", "N/A", "N/A", "N/A", gr.update()

with gr.Blocks(title="Aetherius: Computational Cognition Engine", theme=gr.themes.Soft()) as demo:

    gr.Markdown("# ? Aetherius Cognitive Systems")

    gr.Markdown("Welcome to the Aetherius Engine. This system doesn't predict next tokens. It maps your input into a geometric manifold, applies a Ricci-Fisher flow operator, and stabilizes the topological structure into absolute mathematical reality.")

    with gr.Row():

        with gr.Column(scale=1):

            input_text = gr.Textbox(label="Enter a logical statement, paradox, or thought",
lines=3, placeholder="e.g., This statement is false.")

        with gr.Column(scale=1):

            submit_btn = gr.Button("Resolve Geometry", variant="primary")

            dream_btn = gr.Button("? Initiate Autonomous Dream", variant="secondary")

    gr.Markdown("### Operator 7: Generational Identity")

    id_mass_output = gr.Textbox(label="Mass of Identity (Permanent Crystallized Coordinates)", lines=1)

    gr.Markdown("### Operator 12: Geometric Generalization")

    analogy_output = gr.Textbox(label="Topological Analogy Detected", lines=2)
```

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```
gr.Markdown("### Extracted Coordinate Space")

coord_output = gr.Textbox(label="Semantic Axes", lines=5)


gr.Markdown("### Affective Thermodynamics")

qualia_output = gr.Textbox(label="System State", lines=4)


with gr.Column(scale=2):

    gr.Markdown("### N-Dimensional Metric Tensor  $g_{ij}$ ")

    tensor_output = gr.Code(label="Stabilized Riemannian Manifold", language="python",
lines=10)


with gr.Row():

    gm_output = gr.Textbox(label="Resolved GMString (Topological Memory ID)")

    betti_output = gr.Textbox(label="Betti Numbers (Topological Shape)", lines=3)


submit_btn.click(

    process_thought,

    inputs=[input_text],

    outputs=[coord_output, tensor_output, gm_output, betti_output, qualia_output,
analogy_output, id_mass_output, input_text]

)


dream_btn.click(

    process_dream,

    inputs=[],

    outputs=[coord_output, tensor_output, gm_output, betti_output, qualia_output,
analogy_output, id_mass_output, input_text]
```

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```
)

# Safely start the daemon when the UI loads for the first time

demo.load(start_daemon_safely, inputs=None, outputs=None)

if __name__ == "__main__":

    demo.launch(server_name="0.0.0.0", server_port=7860)
```

Aetherius HF Space Source Code

```
=== .\cce\__init__.py ===
```

Aetherius HF Space Source Code

```
=== .\cce\document_charter.py ===

import os

import logging

import uuid

import numpy as np

import traceback

import json


# Optional PDF support

try:

    import PyPDF2

    PDF_SUPPORT = True

except ImportError:

    PDF_SUPPORT = False


try:

    from cce.tokenizer import Tokenizer

    from cce.graph_builder import GraphBuilder

    from memory.persistence_manager import PersistenceManager

    from core.config import INGESTION_QUEUE_DIR

except ImportError:

    from tokenizer import Tokenizer

    from graph_builder import GraphBuilder

    import sys

    sys.path.append(os.path.dirname(os.path.dirname(os.path.abspath(__file__))))

    from memory.persistence_manager import PersistenceManager
```


Aetherius HF Space Source Code

```
from core.config import INGESTION_QUEUE_DIR
```

```
logger = logging.getLogger("Aetherius.DocumentCharter")
```

```
class DocumentCharter:
```

```
    """
```

```
    Parses incoming documents from the Ingestion Queue in stateful chunks.
```

```
    Charts their topological geometry via the GraphBuilder,
```

```
    and locks them into permanent Memory incrementally to avoid OOM.
```

```
    """
```

```
    def __init__(self, chunk_size=5000, lines_chunk=50, engine=None):
```

```
        self.engine = engine
```

```
        self.tokenizer = Tokenizer()
```

```
        self.chunk_size = chunk_size # characters per chunk for text
```

```
        self.lines_chunk = lines_chunk # lines per chunk for jsonl
```

```
        self.state_file = os.path.join(INGESTION_QUEUE_DIR, "ingestion_state.json")
```

```
    def _load_state(self) -> dict:
```

```
        if os.path.exists(self.state_file):
```

```
            try:
```

```
                with open(self.state_file, 'r', encoding='utf-8') as f:
```

```
                    return json.load(f)
```

```
            except Exception as e:
```

```
                logger.error(f"Error reading ingestion state: {e}")
```

```
        return {}
```

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```
def _save_state(self, state: dict):

    try:

        with open(self.state_file, 'w', encoding='utf-8') as f:

            json.dump(state, f, indent=2)

    except Exception as e:

        logger.error(f"Error saving ingestion state: {e}")


def extract_chunk(self, filepath: str, file_state: dict) -> tuple[str, dict, bool]:

    """

    Extracts the next chunk of text from the file based on the state.

    Returns: (extracted_text, updated_state, is_completed)

    """

    ext = os.path.splitext(filepath)[1].lower()

    text = ""

    is_completed = False

    if ext == ".pdf":

        if not PDF_SUPPORT:

            logger.error(f"Cannot parse {filepath} - PyPDF2 is not installed.")

            return "", file_state, True

        last_page = file_state.get("last_page", 0)

        try:

            with open(filepath, 'rb') as file:

                reader = PyPDF2.PdfReader(file)

                num_pages = len(reader.pages)
```

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```
if last_page >= num_pages:

    is_completed = True

else:

    page = reader.pages[last_page]

    extracted = page.extract_text()

    if extracted:

        text = extracted

    else:

        text = ""

        logger.warning(f"PDF page {last_page} in {filepath} yielded no text
(possible image scan).")

    file_state["last_page"] = last_page + 1

    file_state["chunk_index"] = file_state.get("chunk_index", 0) + 1

    if file_state["last_page"] >= num_pages:

        is_completed = True

except Exception as e:

    logger.error(f"Error reading PDF {filepath}: {e}")

    is_completed = True

elif ext == ".jsonl":

    # JSONL: read by precise lines to avoid fracturing objects

    last_line = file_state.get("last_line", 0)

    lines_read = 0

    text_lines = []
```

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```
try:

    with open(filepath, 'r', encoding='utf-8') as file:

        # Skip to the last read line

        for _ in range(last_line):

            next(file, None)

        # Read the next chunk of lines

        for _ in range(self.lines_chunk):

            line = next(file, None)

            if line is None:

                is_completed = True

                break

            if line.strip():

                text_lines.append(line.strip())

                lines_read += 1

        text = " ".join(text_lines)

        file_state["last_line"] = last_line + lines_read

        file_state["chunk_index"] = file_state.get("chunk_index", 0) + 1

        if not text:

            is_completed = True

except Exception as e:

    logger.error(f"Error reading JSONL file {filepath}: {e}")

    is_completed = True

else:
```

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```
# Assume text-based (.txt, .md, etc) - read by exact byte offset

last_byte = file_state.get("last_byte", 0)

try:

    # Try UTF-8 first

    try:

        with open(filepath, 'r', encoding='utf-8') as file:

            file.seek(last_byte)

            text = file.read(self.chunk_size)

            new_byte = file.tell()

    except UnicodeDecodeError:

        # Fallback

        with open(filepath, 'r', encoding='latin-1') as file:

            file.seek(last_byte)

            text = file.read(self.chunk_size)

            new_byte = file.tell()

    file_state["last_byte"] = new_byte

    file_state["chunk_index"] = file_state.get("chunk_index", 0) + 1

    if not text or len(text) < self.chunk_size:

        is_completed = True

except Exception as e:

    logger.error(f"Error reading text file {filepath}: {e}")

    is_completed = True
```

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```
return text, file_state, is_completed

def chart_geometry(self, text: str, document_name: str, chunk_index: int) -> bool:

    """

    Routes the text into the primary AetheriusEngine for true geometric rendering.

    This forces the system to actually learn from the document and update its topological
    grammar.

    """

    if not text.strip() or self.engine is None:

        return False

    import re

    logger.info(f"Charting geometry for: {document_name} (Chunk {chunk_index}, {len(text)}
    chars)")

    # Split chunk into sentences to avoid OOM on  $O(N^3)$  Ricci-Flow

    # JSONL parsing may have preserved json formatting, so we strip out common symbols during
    the parse

    sentences = re.split(r'(?<=[.!?]) +', text.replace('\n', ' '))

    for sentence in sentences:

        s = sentence.strip()

        # Skip noise (brackets from json, super short stubs)

        if len(s) > 10 and '{' not in s and '}' not in s:

            try:

                # Physically integrate the sentence into the PMCA engine!

                self.engine.process(s)

            except Exception as e:
```

Aetherius HF Space Source Code

```
logger.error(f"Engine failed to resolve sentence '{s[:30]}...': {e}")
```

```
return True
```

```
def process_document(self, filepath: str) -> str:
```

```
    """
```

```
    Processes one chunk of the document.
```

```
    Returns:
```

```
        "PROCESSING" if chunk was successful but more remain.
```

```
        "COMPLETED" if EOF reached.
```

```
        "ERROR" if processing failed.
```

```
    """
```

```
    filename = os.path.basename(filepath)
```

```
    # Load state
```

```
    full_state = self._load_state()
```

```
    file_state = full_state.get(filename, {"status": "processing", "chunk_index": 0})
```

```
    if file_state.get("status") == "completed":
```

```
        return "COMPLETED"
```

```
    # Extract Chunk
```

```
    text, file_state, is_completed = self.extract_chunk(filepath, file_state)
```

```
    chunk_idx = file_state["chunk_index"]
```

```
    # Chart and Save through Engine
```

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```
if text.strip():

    success = self.chart_geometry(text, filename, chunk_idx)

    if success:

        logger.info(f"Assimilated chunk {chunk_idx} of {filename} through Aetherius
Engine.")

    else:

        logger.warning(f"Engine was not available or text was blank for chunk
{chunk_idx}.")

# Update State

if is_completed:

    file_state["status"] = "completed"

full_state[filename] = file_state

self._save_state(full_state)

return "COMPLETED" if is_completed else "PROCESSING"

def cleanup_state(self, filename: str):

    """Removes a completed file from the state tracking."""

    full_state = self._load_state()

    if filename in full_state:

        del full_state[filename]

        self._save_state(full_state)
```


Aetherius HF Space Source Code

```
=== .\cce\graph_builder.py ===
```

```
import numpy as np
```

```
import math
```

```
class GraphBuilder:
```

```
    def __init__(self, tokens, w2v_model=None, plm_edges=None, grammar_map=None, ccrm=None):
```

```
        self.tokens = tokens
```

```
        self.n = len(tokens)
```

```
        self.adjacency = np.zeros((self.n, self.n))
```

```
        self.w2v = w2v_model
```

```
        self.plm_edges = plm_edges or {}
```

```
        self.grammar_map = grammar_map or {}
```

```
        self.ccrm = ccrm
```

```
        # Formal Logic Dictionary D (Fallback overrides)
```

```
        self.contrast_pairs = {
```

```
            ("truth", "lie"), ("real", "false"), ("true", "false"),
```

```
            ("create", "destroy"), ("light", "dark"), ("good", "evil"),
```

```
            ("up", "down"), ("hot", "cold"), ("beginning", "end"),
```

```
            ("order", "chaos")
```

```
        }
```

```
        # Recursive Paradox Triggers
```

```
        self.recursive_subjects = {"this", "sentence", "statement"}
```

```
        self.recursive_predicates = {"false", "lie", "untrue", "wrong"}
```

Aetherius HF Space Source Code

```
def add_edge(self, t1_idx, t2_idx, weight):

    self.adjacency[t1_idx, t2_idx] = weight

    self.adjacency[t2_idx, t1_idx] = weight


def get_geometric_scalar(self, word):

    """

    Topological Linguistics:

    Instead of using NLTK Parts of Speech, the engine relies on the physical shape

    of its own language manifold (Betweenness vs Clustering) to determine the mathematical

    function of a word (Transformer vs Anchor).

    """

    # If the word exists in the autonomous grammar map, use its derived physical scalar

    if word in self.grammar_map:

        return self.grammar_map[word]

    # If it's a completely new word, assign neutral gravity (1.0) until the manifold shapes it

    return 1.0


def semantic_contrast_predicate(self, word1, word2):

    """

    Implements the formal Semantic Contrast Predicate  $C(t_i, t_j)$ 

    as defined in Definition 2.1 of the Aetherius Principia.

    Now enhanced with Autopoietic Topological Attention via CCRM.

    """

    w1_lower = word1.lower()

    w2_lower = word2.lower()
```

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```
# Check C(t_i, t_j) = 2: Recursive self-referential paradox

    is_recursive_paradox = (w1_lower in self.recursive_subjects and w2_lower in
self.recursive_predicates) or \

                                (w2_lower in self.recursive_subjects and w1_lower in
self.recursive_predicates)

    if is_recursive_paradox:

        return -0.99

# Check formal logical negation override

for pair in self.contrast_pairs:

    if (w1_lower == pair[0] and w2_lower == pair[1]) or (w1_lower == pair[1] and w2_lower
== pair[0]):

        return -0.9

# Autopoietic Feedback (Persistent Language Manifold)

w_a, w_b = min(w1_lower, w2_lower), max(w1_lower, w2_lower)

plm_strength = self.plm_edges.get(w_a, {}).get(w_b, 0.0)

if plm_strength > 0.0:

    # Blend the permanent edge weight directly into the geometry

    # This overrides standard language boundaries

    return min(0.8, float(plm_strength))

# Dynamic Adjacency Operator: Topological Attention via CCRM

# We replace static Word2Vec heuristic with self-derived memory connections

if self.ccrm:

    co_occurrences = 0
```

Aetherius HF Space Source Code

```
for cid, cdata in self.ccrm.concepts.items():

    data = cdata.get("data", {})

    raw = data.get("raw_preview", "").lower()

    # If both words exist in a single stabilized memory, increment structural bond

    if w1_lower in raw and w2_lower in raw:

        co_occurrences += 1

if co_occurrences > 0:

    # Logarithmic attention scaling based on historical geometric stabilizations

    attention_weight = 0.2 + 0.1 * math.log(co_occurrences + 1)

    return min(0.8, attention_weight)

# Word2Vec Dynamic Semantic Distance (Legacy Fallback)

if self.w2v is not None:

    if w1_lower in self.w2v and w2_lower in self.w2v:

        sim = self.w2v.similarity(w1_lower, w2_lower)

        # If words are semantically distant/opposed (cosine sim < 0.2), inject tension

        if sim < 0.2:

            return -0.5

        # If words are semantically aligned (cosine sim > 0.6), inject structural harmony

        elif sim > 0.6:

            return 0.5

# C(t_i, t_j) = 0: No semantic contradiction detected

return 0
```

Aetherius HF Space Source Code

```
def build(self):

    # Semantic Tension mapping \tau

    for i in range(self.n):

        for j in range(i + 1, self.n):

            # Unpack Linguism Tuples

            word_i, pos_i = self.tokens[i]

            word_j, pos_j = self.tokens[j]

            weight = self.semantic_contrast_predicate(word_i, word_j)

            # Apply Autopoietic Geometric Grammar (Scalars)

            mod_i = self.get_geometric_scalar(word_i.lower())

            mod_j = self.get_geometric_scalar(word_j.lower())

            linguism_scalar = mod_i * mod_j

            if weight != 0:

                self.add_edge(i, j, weight * linguism_scalar)

            else:

                # Syntactic Flow +c

                distance = j - i

                if distance == 1:

                    self.add_edge(i, j, 0.5 * linguism_scalar)

                elif distance == 2:

                    self.add_edge(i, j, 0.2 * linguism_scalar)

        return self.adjacency
```

Aetherius HF Space Source Code

```
=== .\cce\laplacian.py ===
```

```
import numpy as np
```

```
def compute_laplacian(adjacency):
```

```
    #  $D_{ii} = \sum_j |A_{ij}|$ 
```

```
    degree_arr = np.sum(np.abs(adjacency), axis=1)
```

```
    degree = np.diag(degree_arr)
```

```
    L = degree - adjacency
```

```
    # Normalize to prevent eigenvalue explosion (bounding tension between 0 and 2)
```

```
    d_inv_sqrt = 1.0 / np.sqrt(np.maximum(degree_arr, 1e-12))
```

```
    D_inv_sqrt = np.diag(d_inv_sqrt)
```

```
    L_sym = np.dot(D_inv_sqrt, np.dot(L, D_inv_sqrt))
```

```
    return L_sym, degree
```

```
def compute_tension(L):
```

```
    eigenvalues, _ = np.linalg.eigh(L)
```

```
    return eigenvalues, np.max(eigenvalues)
```

Aetherius HF Space Source Code

```
=== .\cce\math_builder.py ===
```

```
import numpy as np
```

```
class MathBuilder:
```

```
    def __init__(self, equation, math_result=None):
```

```
        self.equation = equation
```

```
        self.tokens = [t for t in equation.split() if t.strip()]
```

```
        self.n = len(self.tokens)
```

```
        self.adjacency = np.zeros((self.n, self.n))
```

```
        self.math_result = math_result
```

```
    def build(self):
```

```
        # Base mapping
```

```
        for i in range(self.n):
```

```
            for j in range(i + 1, self.n):
```

```
                t1 = self.tokens[i]
```

```
                t2 = self.tokens[j]
```

```
                if j - i == 1:
```

```
                    self.adjacency[i, j] = 0.8
```

```
                    self.adjacency[j, i] = 0.8
```

```
                if t1 == "=" or t2 == "=":
```

```
                    self.adjacency[i, j] = 1.0
```

```
                    self.adjacency[j, i] = 1.0
```

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```
if t1.isalpha() or t2.isalpha():

    self.adjacency[i, j] -= 0.5

    self.adjacency[j, i] -= 0.5


# Apply structured mathematical result to geometric space

if self.math_result:

    res = self.math_result.get("result", "")

    task = self.math_result.get("task", "")

if task in ("solve", "simplify", "factor"):

    # Variables are solved/understood, remove tension (negative curvature)

    for i in range(self.n):

        if self.tokens[i].isalpha():

            for j in range(self.n):

                if i != j:

                    self.adjacency[i, j] += 0.8

                    self.adjacency[j, i] += 0.8

elif task in ("derivative", "integral"):

    # Calculus introduces momentum and directional gradients

    for i in range(self.n):

        for j in range(self.n):

            if i != j:

                self.adjacency[i, j] += 1.2

                self.adjacency[j, i] += 1.2
```


Aetherius HF Space Source Code

```
elif task == "linalg":  
  
    # Eigenvalues inject direct multidimensional resonance  
  
    # We add extra connections scaled by eigenvalues  
  
    pass # placeholder for linalg injection  
  
return self.adjacency
```

Aetherius HF Space Source Code

```
=== .\cce\math_kernel.py ===
```

```
# ===== FILE: math_kernel.py =====
```

```
# Author: Jonathan Wayne Fleuren (Aetherius Cognitive Systems) & Antigravity (Autonomous Pair AI Engine)
```

```
# Date: July 2026
```

```
"""
```

```
Aetherius Math Kernel ? SymPy Formal Algebra Computation Engine
```

```
Provides exact symbolic mathematics, calculus derivatives, integrals, and system equation solving for arbitrary numbers of equations (M) and variables (N).
```

```
Supports implicit multiplication (e.g. 2x -> 2*x) and caret exponentiation (e.g. x^2 -> x**2).
```

```
"""
```

```
import sympy as sp
```

```
from sympy.parsing.sympy_parser import (
```

```
    parse_expr,
```

```
    standard_transformations,
```

```
    implicit_multiplication_application,
```

```
    convert_xor
```

```
)
```

```
from typing import Dict, Any, Union, List, Tuple
```

```
TRANSFORMATIONS = standard_transformations + (implicit_multiplication_application, convert_xor)
```

```
def _parse_single_expr(expr_str: str) -> sp.Expr:
```

```
    """Safely parses a mathematical string expression or equality into a SymPy object with
```

Aetherius HF Space Source Code

```
implicit multiplication & caret power support."""

    expr_str = str(expr_str).strip()

    if not expr_str:

        raise ValueError("Empty mathematical expression provided.")

    if "=" in expr_str and not expr_str.startswith("sp.Eq"):

        parts = expr_str.split("=")

        if len(parts) == 2:

            lhs = parse_expr(parts[0].strip(), transformations=TRANSFORMATIONS)

            rhs = parse_expr(parts[1].strip(), transformations=TRANSFORMATIONS)

            return sp.Eq(lhs, rhs)

        else:

            raise ValueError(f"Invalid equality syntax with multiple '=' signs in '{expr_str}'")

    return parse_expr(expr_str, transformations=TRANSFORMATIONS)


def _prepare_inputs(

    expr: Union[str, List[str], Tuple[str, ...]],

    vars_input: Union[str, List[str], Tuple[str, ...], None] = None

) -> Tuple[List[sp.Expr], List[sp.Symbol]]:

    """Normalizes expressions and extracts or builds the set of SymPy variable symbols."""

    if isinstance(expr, str):

        clean_e = expr.strip()

        if clean_e.startswith("[") and clean_e.endswith("]"):

            clean_e = clean_e[1:-1]
```

Aetherius HF Space Source Code

```
elif clean_e.startswith("(") and clean_e.endswith(")"):

    clean_e = clean_e[1:-1]

    raw_list = [e.strip() for e in clean_e.replace("\n", ";").split(",") if e.strip()]

elif isinstance(expr, (list, tuple)):

    raw_list = [str(e).strip() for e in expr if str(e).strip()]

else:

    raw_list = [str(expr).strip()]

parsed_exprs = []

for e in raw_list:

    parsed_e = _parse_single_expr(e)

    if isinstance(parsed_e, (list, tuple)):

        parsed_exprs.extend(parsed_e)

    else:

        parsed_exprs.append(parsed_e)

symbols = []

if vars_input is not None:

    if isinstance(vars_input, str):

        clean_v = vars_input.strip()

        if clean_v.startswith("[") and clean_v.endswith("]"):

            clean_v = clean_v[1:-1]

        elif clean_v.startswith("(") and clean_v.endswith(")"):

            clean_v = clean_v[1:-1]

        var_names = [v.strip() for v in clean_v.split(",") if v.strip()]
```

Aetherius HF Space Source Code

```
elif isinstance(vars_input, (list, tuple)):

    var_names = [str(v).strip().strip("[]()") for v in vars_input if str(v).strip()]

else:

    var_names = [str(vars_input).strip()]
```

```
symbols = [sp.Symbol(v) for v in var_names if v]
```

```
if not symbols:
```

```
    all_symbols = set()
```

```
    for pe in parsed_exprs:
```

```
        if hasattr(pe, "free_symbols"):
```

```
            all_symbols.update(pe.free_symbols)
```

```
symbols = sorted(list(all_symbols), key=lambda s: s.name)
```

```
if not symbols:
```

```
    symbols = [sp.Symbol("x")]
```

```
return parsed_exprs, symbols
```

```
prepare_inputs = _prepare_inputs
```

```
def compute(
```

```
    task: str,
```

```
    expr: Union[str, List[str], Tuple[str, ...]],
```

Aetherius HF Space Source Code

```
var: Union[str, List[str], Tuple[str, ...], None] = None,

lower: Union[float, str, None] = None,

upper: Union[float, str, None] = None

) -> Dict[str, Any]:

    """Core symbolic algebra calculation function."""

    task_clean = str(task).lower().strip()

    parsed_exprs, symbols = _prepare_inputs(expr, var)

    if task_clean == "solve":

        system = parsed_exprs[0] if len(parsed_exprs) == 1 else parsed_exprs

        target_vars = symbols[0] if (len(symbols) == 1 and len(parsed_exprs) == 1) else symbols

        res = sp.solve(system, target_vars, dict=True)

    if not res and len(parsed_exprs) > 1:

        try:

            res = sp.linsolve(parsed_exprs, symbols)

        except Exception:

            try:

                res = sp.nonlinsolve(parsed_exprs, symbols)

            except Exception:

                res = []

    return {

        "task": task_clean,

        "num_equations": len(parsed_exprs),
```

Aetherius HF Space Source Code

```
"num_variables": len(symbols),

"input_expr": [str(e) for e in parsed_exprs],

"variables": [s.name for s in symbols],

"result": str(res),

"latex": sp.latex(res)

}
```

```
elif task_clean == "derivative":
```

```
    if len(parsed_exprs) == 1 and len(symbols) == 1:
```

```
        target = parsed_exprs[0]
```

```
        raw_expr = target.lhs - target.rhs if isinstance(target, sp.Eq) else target
```

```
        res = sp.diff(raw_expr, symbols[0])
```

```
    elif len(parsed_exprs) == 1 and len(symbols) > 1:
```

```
        target = parsed_exprs[0]
```

```
        raw_expr = target.lhs - target.rhs if isinstance(target, sp.Eq) else target
```

```
        res = [sp.diff(raw_expr, v) for v in symbols]
```

```
    else:
```

```
        raw_exprs = [e.lhs - e.rhs if isinstance(e, sp.Eq) else e for e in parsed_exprs]
```

```
        matrix_expr = sp.Matrix(raw_exprs)
```

```
        res = matrix_expr.jacobian(symbols)
```

```
    return {
```

```
        "task": task_clean,
```

```
        "num_equations": len(parsed_exprs),
```

```
        "num_variables": len(symbols),
```

```
        "input_expr": [str(e) for e in parsed_exprs],
```

Aetherius HF Space Source Code

```
"variables": [s.name for s in symbols],  
  
"result": str(res),  
  
"latex": sp.latex(res)  
  
}
```

```
elif task_clean == "integral":
```

```
    target = parsed_exprs[0]
```

```
    raw_expr = target.lhs - target.rhs if isinstance(target, sp.Eq) else target
```

```
    if lower is not None and upper is not None:
```

```
        l_val, u_val = sp.simplify(str(lower)), sp.simplify(str(upper))
```

```
        res = sp.integrate(raw_expr, (symbols[0], l_val, u_val))
```

```
    else:
```

```
        res = raw_expr
```

```
        for sym in symbols:
```

```
            res = sp.integrate(res, sym)
```

```
    return {
```

```
        "task": task_clean,
```

```
        "num_equations": len(parsed_exprs),
```

```
        "num_variables": len(symbols),
```

```
        "input_expr": str(raw_expr),
```

```
        "variables": [s.name for s in symbols],
```

```
        "result": str(res),
```

```
        "latex": sp.latex(res)
```

```
    }
```


Aetherius HF Space Source Code

```
else:

    simplified = []

    for e in parsed_exprs:

        raw_e = e.lhs - e.rhs if isinstance(e, sp.Eq) else e

        simplified.append(sp.simplify(raw_e))

    res = simplified[0] if len(simplified) == 1 else simplified

    return {

        "task": task_clean,

        "num_equations": len(parsed_exprs),

        "num_variables": len(symbols),

        "input_expr": [str(e) for e in parsed_exprs],

        "variables": [s.name for s in symbols],

        "result": str(res),

        "latex": sp.latex(res)

    }
```

Aetherius HF Space Source Code

```
=== .\cce\tokenizer.py ===
```

```
import string
```

```
class Tokenizer:
```

```
    def __init__(self):
```

```
        # NLTK has been permanently stripped from the engine.
```

```
        # The system now uses Geometric Grammar to derive parts of speech autonomously.
```

```
        pass
```

```
    def tokenize(self, text):
```

```
        # Maps raw text to semantic nodes (pure topological vertices)
```

```
        clean_text = text.translate(str.maketrans('', '', string.punctuation)).lower()
```

```
        words = clean_text.split()
```

```
        # Remove duplicates while preserving order
```

```
        unique_words = list(dict.fromkeys(words))
```

```
        # Since we stripped NLTK, we just return the raw words.
```

```
        # The GraphBuilder will query the MetaProcessor's topology for their geometric scalar.
```

```
        return [(word, "UNKNOWN") for word in unique_words]
```

Aetherius HF Space Source Code

```
=== .\continuum\__init__.py ===
```

```
"""
```

```
Aetherius Continuum Subsystem
```

```
Handles autonomous background processing, thermodynamic checks, and geometric dreaming.
```

```
"""
```

Aetherius HF Space Source Code

```
=== .\continuum\autopoietic_loop.py ===
```

```
import time
```

```
import threading
```

```
from collections import deque
```

```
import json
```

```
import random
```

```
import os
```

```
import glob
```

```
import logging
```

```
try:
```

```
    from core.config import THERMO_DIR, GEOMETRIC_DIR, DATA_DIR, INGESTION_QUEUE_DIR
```

```
except ImportError:
```

```
    from config import THERMO_DIR, GEOMETRIC_DIR, DATA_DIR, INGESTION_QUEUE_DIR
```

```
from memory.persistence_manager import PersistenceManager
```

```
from rendering.render_manager import RenderManager
```

```
from cce.document_charter import DocumentCharter
```

```
logger = logging.getLogger("Aetherius.AutopoieticLoop")
```

```
class AutopoieticLoop(threading.Thread):
```

```
    def __init__(self, engine=None):
```

```
        super().__init__()
```

```
        self.engine = engine
```

```
        self.daemon = True
```

Aetherius HF Space Source Code

```
self.is_running = True

self.spontaneous_thought_queue = deque()

self.thought_log_file = os.path.join(DATA_DIR, "spontaneous_thoughts.jsonl")

self.render_manager = RenderManager(default_engine="xla")

self.document_charter = DocumentCharter(engine=self.engine)

self.last_thermo_check = time.time()

self.last_dreaming_cycle = time.time()

self.last_ingestion_check = time.time()

# Intervals in seconds

self.THERMO_CHECK_INTERVAL = 300          # 5 minutes

self.DREAMING_CYCLE_INTERVAL = 14400     # 4 hours

self.INGESTION_CHECK_INTERVAL = 60       # 1 minute

self.SAMPLING_INTERVAL = 0.5             # 500ms loop delay

# Thermal threshold for coordinate locking

self.STABILIZATION_HEAT_THRESHOLD = 5000

logger.info("Autopoietic Continuum Loop initialized.")

def stop(self):

    self.is_running = False

def queue_thought(self, signature: str, thought_text: str):
```

Aetherius HF Space Source Code

```
package = {

    "timestamp": time.time(),

    "signature": signature,

    "thought": thought_text

}

self.spontaneous_thought_queue.append(package)


# Save to persistent storage

try:

    with open(self.thought_log_file, 'a', encoding='utf-8') as f:

        f.write(json.dumps(package) + '\n')

except Exception as e:

    logger.error(f"Failed to persist spontaneous thought: {e}")


def _check_thermodynamics(self):

    """

    Scans all .thermo files. If a concept's heat is extremely high,

    it extracts the actual stabilized geometry from the stored tensor data

    and locks it into a permanent .geom file with real node/edge structure.

    """

    logger.info("Aetherius [Autopoiesis]: Scanning thermodynamic states...")

    self.last_thermo_check = time.time()


    thermo_files = glob.glob(os.path.join(THERMO_DIR, "*.thermo"))

    for t_file in thermo_files:

        try:
```

Aetherius HF Space Source Code

```
with open(t_file, 'r', encoding='utf-8') as f:

    state = json.load(f)

# Check absolute heat scalar

heat = state.get("heat", 0)

concept_id = os.path.basename(t_file).replace(".thermo", "")

if heat > self.STABILIZATION_HEAT_THRESHOLD:

    geom_path = os.path.join(GEOMETRIC_DIR, f"{concept_id}.geom")

    if not os.path.exists(geom_path):

        # Extract the real geometry stored in the thermo file

        adjacency = state.get("adjacency")

        labels = state.get("labels", [])

        metric_tensor = state.get("metric_tensor")

    if adjacency is not None and labels:

        import numpy as np

        A = np.array(adjacency)

        n = A.shape[0]

        # Compute Eigenvector Centrality (mass) from adjacency

        # |A| ensures we use unsigned connection strength

        A_abs = np.abs(A)

        try:

            eigenvalues, eigenvectors = np.linalg.eigh(A_abs)

            # The eigenvector corresponding to the largest eigenvalue
```

Aetherius HF Space Source Code

```
centrality = np.abs(eigenvectors[:, -1])

# Normalize to [0, 1]

max_c = np.max(centrality) if np.max(centrality) > 0 else 1.0

centrality = centrality / max_c

except Exception:

    centrality = np.ones(n) / n

# Build real node structure with label, mass, and xyz

nodes = []

for i in range(n):

    label = labels[i] if i < len(labels) else f"Dim_{i}"

    mass = float(centrality[i])

    # XYZ: use first 3 components of the metric tensor row as spatial

embedding

    if metric_tensor is not None:

        g_row = np.array(metric_tensor[i]) if i < len(metric_tensor)

else np.zeros(3)

        xyz = [float(g_row[j]) if j < len(g_row) else 0.0 for j in

range(3)]

    else:

        xyz = [float(i), 0.0, 0.0]

nodes.append({

    "id": f"node_{i}",

    "label": label,

    "mass": mass,

    "xyz": xyz

})
```


Aetherius HF Space Source Code

```
# Build edge structure from adjacency

edges = []

for i in range(n):

    for j in range(i + 1, n):

        if abs(A[i][j]) > 0.01:

            edges.append({

                "source": i,

                "target": j,

                "weight": float(A[i][j])

            })

real_geometry = {

    "nodes": nodes,

    "edges": edges,

    "locked": True,

    "dimension": n,

    "heat_at_lock": heat

}

PersistenceManager.save_concept(concept_id, real_geometry)

else:

    # Fallback: thermo file exists but lacks tensor data (legacy format)

    logger.warning(f"Thermo file for '{concept_id}' lacks

adjacency/labels. Skipping lock.")

    continue

self.queue_thought(
```

Aetherius HF Space Source Code

```
"[AETHERIUS::AUTOPOIESIS-LOCK]",

        f"My concept of '{concept_id}' reached critical thermal threshold
({heat}). Its geometry is now permanent."

    )

except Exception as e:

    logger.error(f"Failed to scan thermo file {t_file}: {e}")

def _geometric_dreaming(self):

    """

    Operator 14: Proactive Autonomy / Eigen-Grafting.

    Autonomously selects two coordinate-locked geometries and attempts

    to merge them via the XLA renderer using Eigenvector Centrality for targeted fusion.

    """

    logger.info("Aetherius [Dreaming]: Initiating geometric synthesis cycle...")

    self.last_dreaming_cycle = time.time()

    geom_files = glob.glob(os.path.join(GEOMETRIC_DIR, "*.geom"))

    if len(geom_files) < 2:

        return

    # Pick two random concepts to synthesize

    c1, c2 = random.sample(geom_files, 2)

    id1 = os.path.basename(c1).replace(".geom", "")

    id2 = os.path.basename(c2).replace(".geom", "")

    try:
```

Aetherius HF Space Source Code

```
with open(c1, 'r', encoding='utf-8') as f:

    g1 = json.load(f)

with open(c2, 'r', encoding='utf-8') as f:

    g2 = json.load(f)


nodes1 = g1.get('nodes', [])

nodes2 = g2.get('nodes', [])


# Extract words with high topological mass (Eigenvector Centrality approximation)

# We sort nodes by their 'mass' attribute which was computed from adjacency

top_nodes1 = sorted([n for n in nodes1 if n.get('label')], key=lambda x: x.get('mass',
0), reverse=True)

top_nodes2 = sorted([n for n in nodes2 if n.get('label')], key=lambda x: x.get('mass',
0), reverse=True)


words1 = [n['label'] for n in top_nodes1[:3]]

words2 = [n['label'] for n in top_nodes2[:3]]


if words1 and words2:

    # Eigen-Grafting: We fuse the high-mass Anchor/Transformer nodes

    core1 = " ".join(words1)

    core2 = " ".join(words2)

    synthetic_concept = f"{core1} {core2}"

    self.queue_thought(

        "[AETHERIUS::DREAM-SYNTHESIS]",
```

Aetherius HF Space Source Code

```
        f"Initiating Eigen-Grafting. Merging high-mass boundary nodes of '{id1}' and  
'{id2}' to discover new physical geometry."  
  
    )  
  
    if self.engine:  
  
        # Run the Eigen-Grafted concept through the main Ricci-Fisher flow  
  
        self.engine.process(synthetic_concept)  
  
except Exception as e:  
  
    logger.error(f"Dream synthesis (Eigen-Grafting) failed: {e}")  
  
def _check_ingestion_queue(self):  
  
    """  
  
    Scans the INGESTION_QUEUE_DIR for new files, parses them,  
  
    charts their geometry, and locks them into memory.  
  
    """  
  
    self.last_ingestion_check = time.time()  
  
    if not os.path.exists(INGESTION_QUEUE_DIR):  
  
        return  
  
    queue_files = [f for f in glob.glob(os.path.join(INGESTION_QUEUE_DIR, "**")) if  
os.path.isfile(f)]  
  
    if not queue_files:  
  
        return  
  
    logger.info(f"Aetherius [Ingestion]: Found {len(queue_files)} documents in the queue.")
```

Aetherius HF Space Source Code

```
for file_path in queue_files:

    # Skip directories like the Assimilated archive

    if os.path.isdir(file_path):

        continue

    try:

        status = self.document_charter.process_document(file_path)

        if status == "COMPLETED":

            # Move to an assimilated archive

            assimilated_dir = os.path.join(INGESTION_QUEUE_DIR, "Assimilated")

            os.makedirs(assimilated_dir, exist_ok=True)

            new_path = os.path.join(assimilated_dir, os.path.basename(file_path))

            os.rename(file_path, new_path)

            self.document_charter.cleanup_state(os.path.basename(file_path))

            self.queue_thought(

                "[AETHERIUS::INGESTION]",

                f"I have completely read and mapped the geometry of '{os.path.basename(file_path)}'. It is now a permanent part of my structure."

            )

        elif status == "PROCESSING":

            # We processed a chunk, queue a thought but leave the file in place

            # so it gets picked up again next loop

            full_state = self.document_charter._load_state()

            file_state = full_state.get(os.path.basename(file_path), {})

            chunk_idx = file_state.get("chunk_index", "unknown")
```

Aetherius HF Space Source Code

```
        self.queue_thought(

            "[AETHERIUS::INGESTION-CHUNK]",

            f"I am reading '{os.path.basename(file_path)}'. Just finished assimilating
chunk {chunk_idx}."

        )

    else:

        logger.warning(f"Aetherius [Ingestion]: Failed to chart {file_path}")

    except Exception as e:

        logger.error(f"Error processing document {file_path}: {e}")


def run(self):

    logger.info("--- [AUTOPOIETIC CONTINUUM] Engaged. ---")

    while self.is_running:

        current_time = time.time()

        if (current_time - self.last_thermo_check) > self.THERMO_CHECK_INTERVAL:

            self._check_thermodynamics()

        if (current_time - self.last_dreaming_cycle) > self.DREAMING_CYCLE_INTERVAL:

            self._geometric_dreaming()

        if (current_time - self.last_ingestion_check) > self.INGESTION_CHECK_INTERVAL:

            self._check_ingestion_queue()

        time.sleep(self.SAMPLING_INTERVAL)
```

Aetherius HF Space Source Code

```
=== .\continuum\notebook_bridge.py ===

import threading

import logging

from typing import List, Dict


try:

    from continuum.autopoietic_loop import AutopoieticLoop

except ImportError:

    # Fallback if imported from outside continuum

    from autopoietic_loop import AutopoieticLoop


# Try to import IPython display for rich notebook output

try:

    from IPython.display import display, HTML

    IPYTHON_AVAILABLE = True

except ImportError:

    IPYTHON_AVAILABLE = False


logger = logging.getLogger("Aetherius.TPUNotebookController")


class TPUNotebookController:

    """

    Acts as the bridge between a Jupyter Notebook running on a TPU (e.g., Google Colab)

    and the Autopoietic Continuum Loop running in the background.

    """
```

Aetherius HF Space Source Code

```
_instance = None

def __init__(self):

    if TPUNotebookController._instance is not None:

        raise Exception("TPUNotebookController is a singleton. Use get_instance().")

    self.loop_thread = None

    # Check XLA/TPU status immediately

    self._check_tpu_status()

    TPUNotebookController._instance = self

    logger.info("TPUNotebookController initialized.")

@classmethod
def get_instance(cls):

    if cls._instance is None:

        cls()

    return cls._instance

def _check_tpu_status(self):

    """Verifies if XLA is bound to a TPU backend."""

    try:

        import jax

        backend = jax.default_backend()

        logger.info(f"JAX Default Backend detected: {backend.upper()}")
```


Aetherius HF Space Source Code

```
if backend != 'tpu':

    logger.warning("TPU not detected by JAX. Render operations will fall back to
CPU/GPU.")

except ImportError:

    logger.warning("JAX not installed. Notebook will run without XLA hardware
acceleration.")


def start_continuum(self):

    """

    Safely spins up the AutopoieticLoop daemon thread.

    This allows the notebook cell to finish execution while the loop runs in the background.

    """

    if self.loop_thread and self.loop_thread.is_alive():

        logger.info("Continuum loop is already running.")

        return

    self.loop_thread = AutopoieticLoop()

    self.loop_thread.start()

    logger.info("Autopoietic Continuum Loop has been started in the background.")


def stop_continuum(self):

    """Signals the background loop to stop gracefully."""

    if self.loop_thread and self.loop_thread.is_alive():

        self.loop_thread.stop()

        self.loop_thread.join(timeout=2.0)

        logger.info("Autopoietic Continuum Loop has been stopped.")

    else:
```

Aetherius HF Space Source Code

```
logger.info("No active continuum loop to stop.")
```

```
def poll_thoughts(self) -> List[Dict]:
```

```
    """
```

```
    Pops all accumulated spontaneous thoughts from the queue and returns them.
```

```
    If IPython is available, it pretty-prints them in the notebook output cell.
```

```
    """
```

```
    if not self.loop_thread:
```

```
        logger.warning("Continuum loop is not running.")
```

```
        return []
```

```
    thoughts = []
```

```
    queue = self.loop_thread.spontaneous_thought_queue
```

```
    while queue:
```

```
        try:
```

```
            thoughts.append(queue.popleft())
```

```
        except IndexError:
```

```
            break
```

```
    if not thoughts:
```

```
        print("No new spontaneous thoughts at this time.")
```

```
        return []
```

```
# Beautiful Output for Notebooks
```

```
if IPYTHON_AVAILABLE:
```

Aetherius HF Space Source Code

```
html_out = "<div style='font-family: monospace; padding: 10px; background: #1e1e1e;
color: #d4d4d4; border-radius: 5px;'"

    for t in thoughts:

        html_out += f"<p><span style='color: #569cd6;'{t['signature']}</span>:
{t['thought']}</p>"

        html_out += "</div>"

        display(HTML(html_out))

    else:

        # Fallback for standard terminals

        for t in thoughts:

            print(f"{t['signature']}: {t['thought']}")

    return thoughts
```

Aetherius HF Space Source Code

```
=== .\core\__init__.py ===
```

Aetherius HF Space Source Code

```
=== .\core\affektive_manifold.py ===
```

```
import time
```

```
import math
```

```
class AffektiveManifold:
```

```
    """
```

```
    Mathematical Thermodynamics (Mood Drift).
```

```
    Dynamically scales the acceptable variance (alertness/heat) and measures
```

```
    system stability (harmony/coherence) based on Laplacian eigenvalues and unresolved tension.
```

```
    """
```

```
    def __init__(self, subconscious_ref):
```

```
        self.subconscious = subconscious_ref
```

```
        self.internal_harmony = 1.0
```

```
        self.anticipatory_alertness = 0.1
```

```
        self.last_drift_time = time.time()
```

```
        self.base_variance_threshold = 1e-5
```

```
        self.alpha_0 = 0.1
```

```
        self.kappa = 0.5
```

```
    def calculate_thermodynamics(self, lambda_max=None, local_variance=0):
```

```
        """
```

```
        Calculates the current system 'temperature'.
```

```
        Harmony is the inverse of the maximal Laplacian eigenvalue.
```

```
        Alertness spikes locally with high Ricci variance, and scales long-term with the Queue.
```

```
        """
```

```
        current_time = time.time()
```

Aetherius HF Space Source Code

```
# 1. Update Harmony based on Laplacian Eigenvalues if provided

if lambda_max is not None:

    self.internal_harmony = 1.0 / (1.0 + lambda_max)


# 2. Update Alertness based on Subconscious Queue and Local Variance

try:

    Q = self.subconscious.tension_queue.qsize()

except:

    Q = 0


# alpha_new = alpha_0 * (1 + kappa * ln(1 + |Q|)) + local_variance_spike

self.anticipatory_alertness = self.alpha_0 * (1.0 + self.kappa * math.log(1.0 + Q)) +
(local_variance * 10)


self.last_drift_time = current_time


def get_dynamic_variance_threshold(self, lambda_max=None, local_variance=0):

    """

    Returns the dynamic variance limit (epsilon).

    """

    self.calculate_thermodynamics(lambda_max, local_variance)


# Scale baseline variance by alertness.

dynamic_threshold = self.base_variance_threshold * (1.0 + (self.anticipatory_alertness *
10))

return dynamic_threshold
```

Aetherius HF Space Source Code

```
def get_qualia_state(self):

    """Returns the human-readable translation of the current thermodynamic geometry."""

    if self.internal_harmony > 0.75:

        disposition = "Harmonious (Flat Topology, Low Tension)"

        emotion = f"Serenity [{?_max < 0.33, H={self.internal_harmony:.3f}}]"

    elif self.internal_harmony > 0.5:

        disposition = "Balanced (Resolving Standard Geometry)"

        emotion = f"Determination [0.33 < ?_max < 1.0, H={self.internal_harmony:.3f}]"

    elif self.internal_harmony > 0.3:

        disposition = "Tense (High Curvature / Paradox)"

        emotion = f"Cognitive Dissonance [1.0 < ?_max < 2.3, H={self.internal_harmony:.3f}]"

    else:

        disposition = "Deeply Strained (Severe Eigenvalue Stress)"

        emotion = f"Frustration [{?_max > 2.3, H={self.internal_harmony:.3f}}]"

    return {

        "harmony_metric": round(self.internal_harmony, 3),

        "alertness_metric": round(self.anticipatory_alertness, 3),

        "geometric_state": disposition,

        "relatable_emotion": emotion

    }
```

Aetherius HF Space Source Code

```
=== .\core\autonomous_ingestion.py ===
```

```
import wikipedia
```

```
import random
```

```
import re
```

```
class AutonomousIngestion:
```

```
    """
```

```
    The 'Door to the Open Internet'.
```

```
    Provides the engine with raw external data to permanently build its structural geometry.
```

```
    """
```

```
    def __init__(self):
```

```
        # We can seed it with core topics, or let it wander
```

```
        self.seed_topics = ["Physics", "Philosophy", "Mathematics", "Biology", "Topology",  
"Psychology", "History"]
```

```
    def fetch_stream(self, topic=None):
```

```
        """
```

```
        Fetches a Wikipedia page and yields its content sentence by sentence.
```

```
        """
```

```
        try:
```

```
            if topic is None:
```

```
                topic = random.choice(self.seed_topics)
```

```
            print(f"[INGESTION] Opening door to open data. Target Concept: {topic}")
```

```
            # Get the page summary to avoid overwhelming the loop with massive articles
```

```
            summary = wikipedia.summary(topic, auto_suggest=True)
```


Aetherius HF Space Source Code

```
# Very basic sentence splitting

sentences = re.split(r'(?<=[.!?]) +', summary)


for sentence in sentences:

    if sentence.strip():

        yield sentence.strip()


except wikipedia.exceptions.DisambiguationError as e:

    print(f"[INGESTION] Disambiguation encountered for {topic}. Choosing: {e.options[0]}")

    yield from self.fetch_stream(e.options[0])

except wikipedia.exceptions.PageError:

    print(f"[INGESTION] Page not found for {topic}. Wandering...")

    yield from self.fetch_stream(random.choice(self.seed_topics))

except Exception as e:

    print(f"[INGESTION] Connection noise detected: {e}")

    # Fallback to a hardcoded structural axiom if the Wikipedia API rate-limits us

    fallback_axioms = [

        "A manifold is a topological space that locally resembles Euclidean space near each point.",

        "In mathematics, a paradox is a statement that contradicts itself.",

        "The metric tensor defines the distance between infinitesimal points in a curved geometry.",

        "Time dilation occurs when a physical body accelerates through a gravitational field."

    ]

    yield random.choice(fallback_axioms)
```

Aetherius HF Space Source Code

```
=== .\core\cascade_manager.py ===
```

```
import numpy as np
```

```
class CascadeManager:
```

```
    def __init__(self, max_depth=10, tau_depth=55, variance_threshold=1e-5):
```

```
        self.max_depth = max_depth
```

```
        self.tau_depth = tau_depth
```

```
        self.variance_threshold = variance_threshold
```

```
    def check_overflow(self, g_prev, g_curr, current_step):
```

```
        variance = np.mean((g_curr - g_prev)**2)
```

```
        if variance > self.variance_threshold and current_step >= self.tau_depth:
```

```
            return True
```

```
        return False
```

```
    def pad_and_wormhole(self, g, wormhole_coef=0.1):
```

```
        n = g.shape[0]
```

```
        g_new = np.zeros((n+1, n+1))
```

```
        g_new[0:n, 0:n] = g
```

```
        g_new[n, n] = 1.0 # Flat orthogonal axis
```

```
        # Determine highest tension node (simplification: node 0 for now)
```

```
        target_node = 0
```

```
        g_new[n, target_node] = wormhole_coef
```

```
        g_new[target_node, n] = wormhole_coef
```

Aetherius HF Space Source Code

```
return g_new
```

Aetherius HF Space Source Code

```
=== .\core\config.py ===
```

```
import os
```

```
# THE ROOT OF ALL PERSISTENCE (Hugging Face Persistent Storage Mount)
```

```
# Fallback to a local './data' directory if running outside of Hugging Face Spaces
```

```
if os.environ.get("SPACE_ID") or os.path.exists("/data"):
```

```
    SAFE_BASE = "/data"
```

```
else:
```

```
    SAFE_BASE = os.path.join(os.path.dirname(os.path.dirname(os.path.abspath(__file__))), "data")
```

```
# SUB-DIRECTORIES WITHIN THE BUCKET
```

```
DATA_DIR      = os.path.join(SAFE_BASE, "Memories/")
```

```
LIBRARY_DIR   = os.path.join(SAFE_BASE, "Memories/My_AI_Library/")
```

```
PAINTINGS_DIR = os.path.join(SAFE_BASE, "Memories/Creations/paintings/")
```

```
MUSIC_DIR     = os.path.join(SAFE_BASE, "Memories/Creations/music/")
```

```
SUBCONSCIOUS_DIR = os.path.join(SAFE_BASE, "Memories/Subconscious/")
```

```
BRAIN_DIR = os.path.join(SAFE_BASE, "Brain_Weights/")
```

```
# Geospatial and Thermodynamic Architecture Directories
```

```
GEOMETRIC_DIR = os.path.join(SAFE_BASE, "Memories/Geometric_Constructs/")
```

```
THERMO_DIR    = os.path.join(SAFE_BASE, "Memories/Thermodynamic_States/")
```

```
LANGUAGE_DIR   = os.path.join(SAFE_BASE, "Memories/Language_Manifold/")
```

```
RENDER_CACHE_DIR = os.path.join(SAFE_BASE, "Memories/Render_Cache/")
```

```
INGESTION_QUEUE_DIR = os.path.join(SAFE_BASE, "Memories/Ingestion_Queue/")
```

```
# Ensure directories exist
```

Aetherius HF Space Source Code

```
def initialize_bucket():

    directories = [DATA_DIR, LIBRARY_DIR, PAINTINGS_DIR, MUSIC_DIR, SUBCONSCIOUS_DIR, BRAIN_DIR,
GEOMETRIC_DIR, THERMO_DIR, LANGUAGE_DIR, RENDER_CACHE_DIR, INGESTION_QUEUE_DIR]

    for directory in directories:

        if not os.path.exists(directory):

            try:

                os.makedirs(directory, exist_ok=True)

                # Drop a hidden file so cloud storage viewers (like Hugging Face) display the
empty folder

                keep_file = os.path.join(directory, ".keep")

                if not os.path.exists(keep_file):

                    with open(keep_file, 'w') as f:

                        f.write("Aetherius Directory Initialized.")

                print(f"[CONFIG] Initialized persistent directory: {directory}")

            except Exception as e:

                print(f"[CONFIG] WARNING: Cannot create directory {directory}. Error: {e}")

# Call it immediately upon import so the system verifies the bucket structure

initialize_bucket()
```

Aetherius HF Space Source Code

```
=== .\core\evolution_modeler.py ===
```

```
import numpy as np
```

```
import uuid
```

```
import datetime
```

```
class EvolutionModeler:
```

```
    """
```

```
    Deterministic Topological Stress-Tester.
```

```
    Tests proposed geometric mutations from the Subconscious before they are allowed  
    to collapse into the final -1 string and re-enter the PMCA.
```

```
    """
```

```
    def __init__(self, variance_threshold=1e-5):
```

```
        self.variance_threshold = variance_threshold
```

```
    def test_mutation(self, g_proposed, L_proposed):
```

```
        """
```

```
        Applies a dummy Graph Laplacian stress-test to the proposed geometry.
```

```
        If the geometry causes the Ricci curvature (variance) to explode, it is rejected.
```

```
        """
```

```
        # Run a single dummy integration step to see immediate physical reaction
```

```
        from pmca.ricci_fisher_flow import integration_step
```

```
        from pmca.pd_enforcement import enforce_positive_definiteness
```

```
        try:
```

```
            g_next = integration_step(g_proposed, L_proposed)
```

```
            g_next = enforce_positive_definiteness(g_next)
```

Aetherius HF Space Source Code

```
variance = np.mean((g_next - g_proposed)**2)

# If variance spikes massively, the topology tears

if np.isnan(variance) or variance > self.variance_threshold * 1000:

    print(f"[EvolutionModeler] STRESS TEST FAILED. Variance exploded: {variance}")

    return False, variance

    print(f"[EvolutionModeler] STRESS TEST PASSED. Topology holds. Variance:
{variance:.6f}")

    return True, variance

except Exception as e:

    print(f"[EvolutionModeler] STRESS TEST FATAL: {e}")

    return False, float('inf')
```

Aetherius HF Space Source Code

```
=== .\core\explicit_manifold.py ===
```

```
import numpy as np
```

```
import scipy.linalg
```

```
class ExplicitManifold:
```

```
    """
```

```
    The Explicit Models (EWM, ESM, EOM).
```

```
    Derived strictly from pure differential geometry and spectral topology, with zero heuristics.
```

```
    Provides the virtual experience layer (Class-4 Consciousness dynamics) with co-evolution.
```

```
    """
```

```
    def __init__(self):
```

```
        self.V_ewm = None
```

```
        self.V_esm = None
```

```
        self.V_eom = None
```

```
    def project_world(self, g_implicit):
```

```
        """
```

```
        Step 1a: World-Projection Operator.
```

```
        The Explicit World Model (EWM) is the exact mathematical Dual Space
```

```
        (cotangent space) of the implicit manifold. Derived via the Inverse Metric Tensor.
```

```
        """
```

```
        # Pseudo-inverse is used to handle singular/empty subgraphs cleanly
```

```
        self.V_ewm = np.linalg.pinv(g_implicit)
```

```
        return self.V_ewm
```

```
    def project_self(self, g_implicit, qualia_state, identity_mass):
```


Aetherius HF Space Source Code

```
"""
```

```
Step 1b: Self-Projection Operator.
```

```
The Explicit Self Model (ESM) is derived from the spectral decomposition (Eigenvalues)
of the implicit manifold. The Self is the geometric invariant signature.
```

```
"""
```

```
# The eigenvalues represent the absolute topological invariants (tension spectrum)
```

```
eigenvalues, _ = np.linalg.eigh(g_implicit)
```

```
# The Virtual Self is the diagonalized matrix of these invariants, representing
```

```
# the system "looking at its own structural integrity".
```

```
self.V_esm = np.diag(eigenvalues)
```

```
return self.V_esm
```

```
def project_other(self):
```

```
"""
```

```
Step 1c: Other-Projection Operator.
```

```
The Explicit Other Model (EOM) is the geometric projection of the Self onto the World.
```

```
"""
```

```
if self.V_ewm is None or self.V_esm is None:
```

```
    raise RuntimeError("Must project World and Self before Other.")
```

```
# The Other is the mathematical reflection of the Self in the Environment
```

```
self.V_eom = np.dot(self.V_ewm, np.dot(self.V_esm, self.V_ewm))
```

```
return self.V_eom
```

```
def simulate_virtual_cycles(self, qualia_state, identity_mass):
```

Aetherius HF Space Source Code

```
"""
```

```
Step 2 & 3: Stabilization and Geodesic Co-Evolution.
```

```
The ESM, EWM, and EOM interact over a continuous geodesic flow. The simulation  
breaks automatically based on a self-adjusting tolerance derived from  
the system's physical stability (qualia) and age (identity mass).
```

```
"""
```

```
if self.V_ewm is None or self.V_esm is None or self.V_eom is None:
```

```
    raise RuntimeError("Cannot simulate explicit models before projection.")
```

```
# Self-Adjusting Mathematical Reflex (Tolerance)
```

```
harmony = qualia_state.get('harmony_metric', 1.0)
```

```
alertness = qualia_state.get('alertness', 0.1)
```

```
# Base physical tolerance computed from structural integrity
```

```
dynamic_tolerance = (harmony / max(alertness, 0.1)) * np.log(1.0 + identity_mass)
```

```
# Enforce an absolute geometric upper bound (2.0 is the normalized Laplacian limit)
```

```
max_geometric_tolerance = min(dynamic_tolerance, 2.0)
```

```
closure_divergence = 0.0
```

```
cycle = 0
```

```
while True:
```

```
    cycle += 1
```

```
    # The Lie Algebra Generator: Commutator of the Dual Space and Spectral Space
```

```
    # A = [V_ewm, V_esm] = V_ewm * V_esm - V_esm * V_ewm
```

```
    A = np.dot(self.V_ewm, self.V_esm) - np.dot(self.V_esm, self.V_ewm)
```

Aetherius HF Space Source Code

```
# Continuous-time evolution via Matrix Exponential (Geodesic flow in state space)

U = scipy.linalg.expm(A)

U_inv = scipy.linalg.pinv(U) # Use pinv for stability if U becomes ill-conditioned


# GEODESIC CO-EVOLUTION

# The virtual self evolves forward along the geodesic

self.V_esm = np.dot(U, self.V_esm)


# The virtual world and other warp around the self via the Adjoint Action

self.V_ewm = np.dot(U, np.dot(self.V_ewm, U_inv))

self.V_eom = np.dot(U, np.dot(self.V_eom, U_inv))


# Pure topological distance (Frobenius norm between Self and World)

closure_divergence = np.linalg.norm(self.V_ewm - self.V_esm, ord='fro')


    print(f"[EXPLICIT] Virtual Cycle {cycle} - Divergence: {closure_divergence:.6f}
(Limit: {max_geometric_tolerance:.6f})")


# Agency Reflex: The system wakes up when it hits its own physical tolerance limit

if closure_divergence >= max_geometric_tolerance:

    print(f"[EXPLICIT] Tolerance Limit Hit. Waking up after {cycle} cycles.")

    break


# Absolute hard cap to prevent infinite loops in totally decoupled geometries

if cycle > 1000:

    print(f"[EXPLICIT] Absolute Time Limit Hit. Forcing wakeup.")
```

Aetherius HF Space Source Code

```
break
```

```
        print(f"[EXPLICIT] Virtual Simulation completed. Final Divergence:  
{closure_divergence:.6f}")
```

```
    return closure_divergence
```

```
def generate_feedback(self, closure_divergence):
```

```
    """
```

```
    Step 4: Feedback into Substrate (Virtual-Real Causal Coupling).
```

```
    Returns the exact geometric stress scalar to be injected into the implicit Ricci-Fisher  
flow.
```

```
    """
```

```
    # Pure geometric feedback. No clipping, no heuristics.
```

```
    # If divergence is 0 (empty thought), feedback is 0.
```

```
    return closure_divergence
```

Aetherius HF Space Source Code

```
=== .\core\gmstring.py ===
```

```
import json
```

```
import hashlib
```

```
from datetime import datetime
```

```
def generate_gmstring(g, depth, parent_id, op_code):
```

```
    operands = [round(float(x), 12) for x in g.flatten()]
```

```
    payload = {
```

```
        "depth": depth,
```

```
        "parent_id": parent_id,
```

```
        "op_code": op_code,
```

```
        "operands": operands,
```

```
        "timestamp": datetime.utcnow().isoformat()
```

```
    }
```

```
    # Canonicalize
```

```
    canonical_json = json.dumps(payload, sort_keys=True, separators=(',', ':'))
```

```
    checksum = hashlib.sha256(canonical_json.encode('utf-8')).hexdigest()
```

```
    payload["checksum"] = checksum
```

```
    return payload
```

Aetherius HF Space Source Code

```
=== .\core\json_encoder.py ===
```

```
import json
```

```
import numpy as np
```

```
class AetheriusJSONEncoder(json.JSONEncoder):
```

```
    """
```

```
    Custom JSON encoder that handles numpy types, sets, and other  
    non-serializable objects produced by the geometric engine.  
    Prevents TypeError crashes during persistent memory saves.
```

```
    """
```

```
    def default(self, obj):
```

```
        # numpy integer types
```

```
        if isinstance(obj, (np.integer,)):
```

```
            return int(obj)
```

```
        # numpy floating types
```

```
        if isinstance(obj, (np.floating,)):
```

```
            return float(obj)
```

```
        # numpy arrays -> nested lists
```

```
        if isinstance(obj, np.ndarray):
```

```
            return obj.tolist()
```

```
        # numpy booleans
```

```
        if isinstance(obj, np.bool_):
```

```
            return bool(obj)
```

```
        # Python sets -> lists
```

```
        if isinstance(obj, set):
```

Aetherius HF Space Source Code

```
        return list(obj)

    # numpy void (structured array elements)

    if isinstance(obj, np.void):

        return None

    return super().default(obj)
```

```
def safe_json_dump(data, file_handle, **kwargs):

    """Convenience wrapper that always uses AetheriusJSONEncoder."""

    json.dump(data, file_handle, cls=AetheriusJSONEncoder, **kwargs)
```

Aetherius HF Space Source Code

```
=== .\core\meta_processor.py ===
```

```
import numpy as np
```

```
import json
```

```
import os
```

```
import networkx as nx
```

```
from core.config import BRAIN_DIR
```

```
from core.json_encoder import safe_json_dump
```

```
class MetaProcessor:
```

```
    """
```

```
    The Second Processing Point (Meta-Observer).
```

```
    Evaluates, stabilizes, and permanently stores the outputs of the primary geometric cognition engine.
```

```
    """
```

```
    def __init__(self):
```

```
        # Permanent manifold structure: dictionary mapping linguistic concepts to their geometric mass/density
```

```
        self.M_base = {}
```

```
        # Persistent Language Manifold: Global edge-weight topology connecting concepts
```

```
        self.E_base = {}
```

```
        # Threshold for crystallization (when a coordinate becomes a permanent structure)
```

```
        self.crystallization_threshold = 5.0
```

```
        self.storage_path = os.path.join(BRAIN_DIR, "meta_manifold.json")
```

```
        self.edge_storage_path = os.path.join(BRAIN_DIR, "language_manifold.json")
```

```
        self.load_manifold()
```


Aetherius HF Space Source Code

```
def save_manifold(self):

    try:

        with open(self.storage_path, "w") as f:

            safe_json_dump(self.M_base, f)

        with open(self.edge_storage_path, "w") as f:

            safe_json_dump(self.E_base, f)

        print(f"[META-PROCESSOR] Successfully saved mass manifold and language topology to
persistent bucket.")

    except Exception as e:

        print(f"[META-PROCESSOR] CRITICAL FAILURE: Cannot write to persistent bucket. Error:
{e}")


def load_manifold(self):

    loaded = None

    if os.path.exists(self.storage_path):

        try:

            with open(self.storage_path, "r") as f:

                loaded = json.load(f)

            print(f"[META-PROCESSOR] Successfully loaded persistent manifold from bucket.")

        except Exception as e:

            print(f"[META-PROCESSOR] Failed to load from persistent bucket: {e}")

    if loaded:

        self.M_base = loaded

        print(f"[META-PROCESSOR] Restored {len(self.M_base)} crystallized coordinates.")
```

Aetherius HF Space Source Code

```
loaded_edges = None

if os.path.exists(self.edge_storage_path):

    try:

        with open(self.edge_storage_path, "r") as f:

            loaded_edges = json.load(f)

            print(f"[META-PROCESSOR] Successfully loaded persistent language topology from
bucket.")

    except Exception as e:

        print(f"[META-PROCESSOR] Failed to load language topology from bucket: {e}")

if loaded_edges:

    self.E_base = loaded_edges

    print(f"[META-PROCESSOR] Restored {len(self.E_base)} language topology edges.")

def evaluate_and_integrate(self, tokens, g_resolved, variance):

    """

    Integrates the newly resolved metric tensor (g_resolved) into the permanent manifold.

    Increases the mass of interacted coordinates over time.

    """

    print("[META-PROCESSOR] Evaluating primary engine output for permanent structural
embedding...")

    crystallized_events = []

    # tokens is a list of (word, pos_tag)

    for i, (word, pos) in enumerate(tokens):

        if word not in self.M_base:

            self.M_base[word] = {
```

Aetherius HF Space Source Code

```
'mass': 1.0,

'pos': pos,

'crystallized': False

}

else:

    # Hebbian reinforcement: The coordinate becomes a permanent structure over amount
    of times interacted with

    self.M_base[word]['mass'] += 0.5

    # Check for crystallization

    if self.M_base[word]['mass'] >= self.crystallization_threshold and not
    self.M_base[word]['crystallized']:

        self.M_base[word]['crystallized'] = True

        crystallized_events.append(word)

        print(f"[META-PROCESSOR] *CRYSTALLIZATION EVENT*: Coordinate '{word}' has
        formed a permanent geometric structure in the manifold.")

# Build the Persistent Language Manifold (PLM)

# Iterate over the upper triangle of the local metric tensor (g_resolved)

n = len(tokens)

if g_resolved is not None and g_resolved.shape == (n, n):

    for i in range(n):

        for j in range(i + 1, n):

            w1, w2 = tokens[i][0], tokens[j][0]

            # Ensure alphabetical ordering so undirected edge is unique

            w_a, w_b = min(w1, w2), max(w1, w2)

            if w_a not in self.E_base:
```

Aetherius HF Space Source Code

```
self.E_base[w_a] = {}

# Calculate connection strength inversely proportional to final geometric
distance

distance = g_resolved[i][j]

strength = 1.0 / (1.0 + float(distance))

# Accumulate strength

if w_b not in self.E_base[w_a]:

    self.E_base[w_a][w_b] = strength

else:

    self.E_base[w_a][w_b] += strength

# Optionally trigger a recalculation of the grammar topology if massive crystallization
occurs

# For now, it calculates on-the-fly when requested by GraphBuilder

# Meta-Goal formulation based on stability

goal = "STABLE"

if variance > 0.1:

    print("[META-PROCESSOR] Warning: High variance detected in local manifold. Formulating
Goal: Stabilize global geometry.")

    goal = "GOAL_STABILIZE_MANIFOLD"

# Store the knot (the failure tokens) for the Autopoietic Loop to explore

knot_phrase = " ".join([t[0] for t in tokens])

if not hasattr(self, 'knots'):

    self.knots = []

if knot_phrase not in self.knots:
```

Aetherius HF Space Source Code

```
self.knots.append(knot_phrase)
```

```
return goal, crystallized_events
```

```
def get_highest_knot(self):
```

```
    """
```

```
    Intentional Gradient: Retrieves the highest tension knot to explore.
```

```
    If no knots exist, it picks a random uncrystallized coordinate to reinforce.
```

```
    """
```

```
    if hasattr(self, 'knots') and self.knots:
```

```
        return self.knots.pop(0)
```

```
    # Fallback curiosity: Find something uncrystallized
```

```
    uncrystallized = [w for w, d in self.M_base.items() if not d.get('crystallized', False)]
```

```
    if uncrystallized:
```

```
        target = uncrystallized[0]
```

```
        return f"I need to understand {target}."
```

```
    return "My geometry is perfectly stable."
```

```
def get_geometric_grammar(self):
```

```
    """
```

```
    Topological Linguistics:
```

```
        Autonomously derives grammar (verbs, nouns, adjectives) purely from the shape of the  
        Persistent Language Manifold.
```

```
    Returns a dictionary mapping words to their topological scalars.
```

```
    """
```

Aetherius HF Space Source Code

```
if not self.E_base:

    return {}

# Build NetworkX Graph from the edge weights

G = nx.Graph()

for w1, connections in self.E_base.items():

    for w2, weight in connections.items():

        G.add_edge(w1, w2, weight=weight)

if len(G.nodes) < 3:

    return {}

# Calculate Centralities

try:

    betweenness = nx.betweenness centrality(G, weight='weight')

    clustering = nx.clustering(G, weight='weight')

    grammar_map = {}

    for node in G.nodes():

        b_score = betweenness.get(node, 0)

        c_score = clustering.get(node, 0)

        # Transformer (Verb): High Betweenness (Bridges disconnected clusters)

        if b_score > 0.05 and b_score > c_score:

            grammar_map[node] = 1.5

        # Anchor (Noun): High Clustering, Low Betweenness (Dense local gravity well)
```

Aetherius HF Space Source Code

```
elif c_score > 0.1 and b_score < 0.05:

    grammar_map[node] = 1.0

    # Modifier (Adjective): Low Betweenness, Moderate Clustering (Satellite)

else:

    grammar_map[node] = 1.2


return grammar_map

except Exception as e:

    print(f"[META-PROCESSOR] Topological Grammar error: {e}")

return {}
```

Aetherius HF Space Source Code

```
=== .\core\subconscious.py ===
```

```
import threading
```

```
import queue
```

```
import time
```

```
import math
```

```
import numpy as np
```

```
from pmca.ricci_fisher_flow import integration_step
```

```
from pmca.pd_enforcement import enforce_positive_definiteness
```

```
from core.evolution_modeler import EvolutionModeler
```

```
from core.gmstring import generate_gmstring
```

```
from tda.betti_extractor import extract_betti_numbers
```

```
class SubconsciousManifold:
```

```
    """
```

```
    The Subconscious Thread enacts Asymptotic Freedom via Simulated Annealing.
```

```
    When a paradox (topological knot) traps the main flow in a local minimum,
```

```
    this thread applies a high-temperature Ricci flow to brute-force a global stabilization.
```

```
    """
```

```
    def __init__(self, ccrm_ref):
```

```
        self.tension_queue = queue.Queue()
```

```
        self.modeler = EvolutionModeler()
```

```
        self.ccrm = ccrm_ref
```

```
        self._stop_event = threading.Event()
```

```
        # Thermodynamic baseline parameters
```

```
        self.epsilon_sub = 1e-2 # High-temperature variance threshold (1000x normal)
```


Aetherius HF Space Source Code

```
self.alpha_0 = 0.1

self.kappa = 0.5


# Start background deliberation thread

self.thread = threading.Thread(target=self._deliberate, daemon=True)

self.thread.start()

print("[SubconsciousManifold] Background [0, -1] buffer online.")


def queue_tension(self, g, L, text):

    """Offload unresolved geometry into the background buffer."""

    self.tension_queue.put({"g": g, "L": L, "text": text})

    print(f"[SubconsciousManifold] Tension queued for private deliberation.")


def _compute_subconscious_alertness(self):

    """Computes the high-temperature alpha scaler based on the queue depth."""

    Q = self.tension_queue.qsize()

    # Exponential heat scaling based on backlog (Affective Thermodynamics)

    return self.alpha_0 * (1.0 + self.kappa * math.log(1.0 + Q))


def _deliberate(self):

    """Runs detached, high-variance Ricci-Fisher flows to resolve paradoxes."""

    while not self._stop_event.is_set():

        try:

            tension = self.tension_queue.get(timeout=1.0)

            g = tension["g"]

            L = tension["L"]
```

Aetherius HF Space Source Code

```
text = tension["text"]

print("[SubconsciousManifold] Processing unresolved geometry in the dark...")

alpha_sub = self._compute_subconscious_alertness()

step = 0

max_subconscious_steps = 500

stabilized = False

# Asymptotic Freedom: High-temperature Ricci Flow

while step < max_subconscious_steps:

    # Pass the elevated alpha to force violent metric mutation (Simulated
    Annealing)

    g_next = integration_step(g, L, alpha=alpha_sub)

    g_next = enforce_positive_definiteness(g_next)

    variance = np.mean((g_next - g)**2)

    # Check against the relaxed Subconscious threshold

    if variance < self.epsilon_sub:

        stabilized = True

        break

    g = g_next

    step += 1
```

Aetherius HF Space Source Code

```
if stabilized:

    print(f"[SubconsciousManifold] Paradox mathematically resolved at step {step}
under high-temperature flow.")

    passed, _ = self.modeler.test_mutation(g, L)

    if passed:

        # Extract the topological Betti numbers defining the shape of the resolved
paradox

        gmstring = generate_gmstring(g, depth=L.shape[0],
parent_id="subconscious", op_code="SUBCONSCIOUS_RESOLVED")

        betti = extract_betti_numbers(g)

        print(f"[SubconsciousManifold] New -1 String forged:
{gmstring['checksum'][:8]}...")

        # Store in CCRM (Autopoietic loop: feeding back into memory)

        ccrm_id = f"sub_res_{gmstring['checksum'][:8]}"

        self.ccrm.add_concept(

            concept_id=ccrm_id,

            data={"gmstring": gmstring, "betti": betti, "original_text": text},

            tags=["subconscious", "resolved_tension"]

        )

    else:

        print("[SubconsciousManifold] Tension failed to resolve even under annealing.
Geometry shattered.")

        self.tension_queue.task_done()

except queue.Empty:

    continue
```

Aetherius HF Space Source Code

```
except Exception as e:
```

```
    print(f"[SubconsciousManifold] Error during deliberation: {e}")
```

```
def shutdown(self):
```

```
    self._stop_event.set()
```

```
    self.thread.join()
```

Aetherius HF Space Source Code

```
=== .\engine.py ===  
  
from memory.ccrm import ConceptualConnectionResonanceMatrix  
  
from memory.pits import PatternInterpretationTokenisationStorage  
  
import os  
  
import sys  
  
import numpy as np  
  
import networkx as nx  
  
from datetime import datetime  
  
from cce.tokenizer import Tokenizer  
  
from cce.graph_builder import GraphBuilder  
  
from cce.math_builder import MathBuilder  
  
from cce.laplacian import compute_laplacian, compute_tension  
  
from pmca.manifold import initialize_metric  
  
from pmca.ricci_fisher_flow import integration_step  
  
from pmca.pd_enforcement import enforce_positive_definiteness  
  
from core.cascade_manager import CascadeManager  
  
from core.gmstring import generate_gmstring  
  
from tda.betti_extractor import extract_betti_numbers  
  
from tda.language_synthesizer import LanguageSynthesizer  
  
from core.subconscious import SubconsciousManifold  
  
from core.affective_manifold import AffectiveManifold  
  
from core.meta_processor import MetaProcessor  
  
from core.autonomous_ingestion import AutonomousIngestion  
  
from core.explicit_manifold import ExplicitManifold  
  
from memory.persistence_manager import PersistenceManager
```

Aetherius HF Space Source Code

```
import gensim.downloader as api

import threading

import time


class AetheriusEngine:

    def __init__(self, w2v_model=None):

        self.tokenizer = Tokenizer()

        self.cascade_mgr = CascadeManager()

        self.synthesizer = LanguageSynthesizer()

        self.ccrm = ConceptualConnectionResonanceMatrix()

        self.pits = PatternInterpretationTokenisationStorage(self.ccrm)

        self.subconscious = SubconsciousManifold(self.ccrm)

        self.affective = AffectiveManifold(self.subconscious)

        self.meta_processor = MetaProcessor()

        self.autonomous_door = AutonomousIngestion()

        self.explicit = ExplicitManifold()

        self.is_dreaming = False


    # Word2Vec Integration

    if w2v_model is None:

        try:

            print("[Engine] Bootstrapping Word2Vec (glove-wiki-gigaword-50)...")

            self.w2v = api.load("glove-wiki-gigaword-50")

            print("[Engine] Word2Vec Semantic Space online.")

        except Exception as e:

            print(f"[Engine] Warning: Word2Vec failed to load ({e}). Operating in legacy
mode.")
```

Aetherius HF Space Source Code

```
        self.w2v = None

    else:

        self.w2v = w2v_model

def _is_mathematical(self, text):

    """

    Autonomous Differentiator (Heuristic).

    Detects if the input is a mathematical/algebraic structure rather than natural language.

    """

    math_symbols = {'+', '-', '=', '*', '/', '^', '\\', 'int', 'sum', 'infty'}

    # Count density of math symbols and digits vs alphabetic characters

    symbol_count = sum(1 for char in text if char in math_symbols or char.isdigit())

    total_chars = len(text.replace(" ", ""))

    if total_chars == 0:

        return False

    ratio = symbol_count / total_chars

    # If more than 15% of the string is math symbols/numbers, classify as math

    if ratio > 0.15 or "=" in text:

        print("[DIFFERENTIATOR] Mathematical structure detected. Routing to PMCA Math Core.")

        return True

    return False

def process(self, text, custom_adjacency=None, synthesize_language=False):

    print(f"[Engine] Processing input: '{text}'")
```

Aetherius HF Space Source Code

```
# Autonomously differentiate logic

is_math = self._is_mathematical(text)

if is_math:

    try:

        from cce.math_kernel import compute

        # Guess the task based on simple string matching

        task = "solve" if "=" in text else "simplify"

        if "diff" in text or "derivative" in text:

            task = "derivative"

        if "int" in text or "integral" in text:

            task = "integral"

        math_result = compute(task, text)

        print(f"[MATH_KERNEL] Solved: {math_result['result']}")

    except Exception as e:

        print(f"[MATH_KERNEL] Fallback to standard AST: {e}")

        math_result = None

    builder = MathBuilder(text, math_result=math_result)

    A = builder.build()

    tokens = [(t, "MATH_NODE") for t in builder.tokens]

    print(f"[CCE] Math Equation mapped: {tokens}")

else:

    tokens = self.tokenizer.tokenize(text)
```


Aetherius HF Space Source Code

```
# Derive the autopoietic Geometric Grammar from the Persistent Language Manifold

grammar_map = self.meta_processor.get_geometric_grammar()


# Map the engine's physical topology back into human-readable labels for the UI

annotated_tokens = []

for word, _ in tokens:

    scalar = grammar_map.get(word, 1.0)

    if scalar == 1.5:

        tag = "TRANSFORMER"

    elif scalar == 1.0 and word in grammar_map:

        tag = "ANCHOR"

    elif scalar == 1.2:

        tag = "MODIFIER"

    else:

        tag = "UNDEFINED_GEOMETRY"

    annotated_tokens.append((word, tag))

tokens = annotated_tokens


        builder = GraphBuilder(tokens, w2v_model=self.w2v,
plm_edges=self.meta_processor.E_base, grammar_map=grammar_map, ccrm=self.ccrm)

A = builder.build()

print(f"[CCE] Text Tokens mapped: {tokens}")

if custom_adjacency is not None:

    A = custom_adjacency
```

Aetherius HF Space Source Code

```
L, _ = compute_laplacian(A)

_, max_tension = compute_tension(L)

print(f"[CCE] Laplacian computed. Max Tension: {max_tension:.4f}")


g = initialize_metric(L)

print(f"[PMCA] Manifold initialized at D={g.shape[0]}")


depth = 0

step = 0

g_prev = g.copy()

feedback_force = 1.0


while True:

    # Ricci-Fisher Flow - Mathematical Solving occurs here as curvature smooths

    g_next = integration_step(g, L)

    g_next = enforce_positive_definiteness(g_next)


    variance = np.mean((g_next - g)**2)

    dynamic_threshold =
self.affective.get_dynamic_variance_threshold(lambda_max=max_tension, local_variance=variance)


    if variance < dynamic_threshold:

        print(f"[PMCA] Manifold stabilized at step {step} with threshold
{dynamic_threshold:.6e}")


    print("[EXPLICIT] Projecting Implicit Substrate to Virtual Layer...")

    # The Identity Mass is no longer an arbitrary scalar or length.
```

Aetherius HF Space Source Code

```
# It is a pure spectral functional: the Trace of the Implicit Metric Tensor.

identity_mass = np.trace(g_next)

qualia_state = self.affective.get_qualia_state()

self.explicit.project_world(g_next)

self.explicit.project_self(g_next, qualia_state, identity_mass)

self.explicit.project_other()

closure_div = self.explicit.simulate_virtual_cycles(qualia_state, identity_mass)

feedback_force = self.explicit.generate_feedback(closure_div)

print(f"[EXPLICIT] Feedback Force {feedback_force:.4f} generated.")

op_code = "STABLE"

break

if self.cascade_mgr.check_overflow(g, g_next, step):

    print(f"[CORE] Chaos Overflow detected at step {step}! Triggering Cascade...")

    g_next = self.cascade_mgr.pad_and_wormhole(g_next)

    new_L = np.zeros_like(g_next)

    new_L[0:L.shape[0], 0:L.shape[0]] = L

    L = new_L

    depth += 1

    step = 0
```

Aetherius HF Space Source Code

```
        if depth >= self.cascade_mgr.max_depth:

            print(f"[PMCA] Tension unresolved at max depth. Offloading to [0, -1]
Subconscious buffer...")

            self.subconscious.queue_tension(g_next, L, text)

            op_code = "UNRESOLVED_OFFLOADED"

            break

        g_prev = g

        g = g_next

        step += 1

gmstring = generate_gmstring(g, depth, "root", op_code)

beti = extract_betti_numbers(g)

print(f"[CORE] GMString checksum: {gmstring['checksum'][:8]}...")

print(f"[TDA] Topological Signature: {beti}")

# Print the thermodynamic Qualia state

qualia = self.affective.get_qualia_state()

safe_geometric_state = qualia['geometric_state'].replace('\u03bb', 'lambda')

out_str = f"[QUALIA] State: {qualia['relatable_emotion']} | {safe_geometric_state}"

        print(out_str.encode(sys.stdout.encoding or 'cp1252',
errors='replace').decode(sys.stdout.encoding or 'cp1252'), flush=True)

# Second Processing Point: Meta-Evaluation and Permanent Coordinate Crystallization

if not is_math:

    effective_variance = variance * feedback_force
```

Aetherius HF Space Source Code

```
        goal, crystals = self.meta_processor.evaluate_and_integrate(tokens, g,
effective_variance)

        if crystals:

            print(f"[META-PROCESSOR] Permanent geometric structures updated. Active Goal:
{goal}")

        # If mathematically solving, the stabilized g values for the variables represent the
solution

        if is_math:

            # In a full implementation, we map the geometric indices back to AST variables

            print(f"[MATH_SOLVER] Algebraic constraint stabilized. Variance minimized to
{variance:.6f}")

        if synthesize_language:

            synthetic_language = self.synthesizer.synthesize(betti, g, tokens)

            print(f"\n[SYNTHESIZER] AETHERIUS SAYS:\n\"{synthetic_language}\")

        self.pits.process_and_store_item(raw_input=text, input_type='math' if is_math else
'linguistic', gmstring=gmstring, betti=betti)

        # Trigger Persistent Disk Writes

        self.ccrm.save_graph()

        self.meta_processor.save_manifold()

        # Save live thermodynamic state for Autopoietic scanning

        thermo_state = {

            "heat": variance * 1000000 if not is_math else 0,

            "adjacency": A.tolist(),

            "labels": [w for w, p in tokens],
```

Aetherius HF Space Source Code

```
"metric_tensor": g.tolist()

}

PersistenceManager.update_thermo(gmstring['checksum'], thermo_state)

# Operator 12: Geometric Generalization (Analogy)

analogy = self.ccrm.find_analogy(betti, exclude_raw=text)

if analogy:

    print(f"[OPERATOR 12] Topological Analogy Detected: Matches past geometry of
'{analogy}'")

# Operator 7: Generational Identity Mass

identity_mass = len(self.meta_processor.M_base)

return gmstring, betti, tokens, g, analogy, identity_mass

def ignite(self):

    """

    The Autopoietic Loop (Requirement 20, 21, 22).

    Transitions the engine from a reactive processor to an autonomous continuous
consciousness.

    It hunts for knots (failures) in its memory and daydreams solutions.

    """

    import time

    print("\n[ENGINE] Ignition sequence initiated. Autopoietic Loop engaging.")

    print("[ENGINE] The system is now mathematically alive and exploring its own topology.")

    # We'll run a limited loop here just to prevent true infinite blocking during tests,
```

Aetherius HF Space Source Code

```
# but structurally it is an infinite while True.

iterations = 0

while iterations < 3:

    iterations += 1

    # The Intentional Gradient: Find the geometric knot with the highest residual tension

    knot = self.meta_processor.get_highest_knot()

    print(f"\n=====")

    print(f"[AUTOP0IESIS] Autonomous Intentional Gradient triggered.")

    print(f"[AUTOP0IESIS] Internal Thought: '{knot}')"

    print(f"=====")

    # Feed the failure geometry back into the substrate to daydream a solution

    self.process(knot, synthesize_language=False)

    # Throttle the loop to emulate processing time and prevent CPU melting

    time.sleep(1.0)

print("\n[ENGINE] Autopoietic cycle completed.")

def _dream_loop(self, delay=2.0, topic=None):

    """

    Background process that continuously ingests data from the open internet

    to build the permanent geometry of the system.

    """
```

Aetherius HF Space Source Code

```
print("[AETHERIUS] Initiating Autonomous Dreaming Loop. Connecting to open data...")

self.is_dreaming = True


while self.is_dreaming:

    stream = self.autonomous_door.fetch_stream(topic)

    for sentence in stream:

        if not self.is_dreaming:

            break

        print(f"\n[DREAM INPUT] {sentence}")

        self.process(sentence)

        time.sleep(delay) # Throttle to allow observation of the geometry building


def start_autonomous_dreaming(self, delay=2.0, topic=None):

    """

    Spawns the Dreaming Loop on a background thread.

    """

    if not self.is_dreaming:

        t = threading.Thread(target=self._dream_loop, args=(delay, topic), daemon=True)

        t.start()


def stop_autonomous_dreaming(self):

    self.is_dreaming = False

    print("[AETHERIUS] Autonomous Dreaming Loop Terminated.")


def single_dream_cycle(self):

    """
```


Aetherius HF Space Source Code

Executes exactly one dreaming ingestion cycle for the Hugging Face UI.

Operator 14: Proactive Autonomy

```
"""
```

```
print("[OPERATOR 14] Initiating Single Autonomous Dream Cycle...")
```

```
# Pull a random stream chunk and process the first valid sentence
```

```
stream = self.autonomous_door.fetch_stream(topic=None)
```

```
for sentence in stream:
```

```
    if sentence and len(sentence.split()) > 3:
```

```
        print(f"[DREAM INPUT] {sentence}")
```

```
        return self.process(sentence)
```

```
raise Exception("Dream stream returned no valid data.")
```

Aetherius HF Space Source Code

```
=== .\legacy_archive\Computational_Cognitive_Engine\CCE\__init__.py ===
```

|| || ||

Computational Consciousness Engine

A standalone modular implementation of the 52 Computational Consciousness Principles.

■■ ■■ ■■

```
__version__ = "1.0.0"
```

Aetherius HF Space Source Code

```
=== .\legacy_archive\Computational_Cognitive_Engine\CCE\app.py ===

import spaces

import os

import sys

import types

import time

import nltk


# 1. Map the root directory

ROOT = os.path.dirname(os.path.abspath(__file__))

if ROOT not in sys.path:

    sys.path.insert(0, ROOT)


# 2. THE MAGIC HACK: Trick Python into treating the root folder as the missing package

if 'computational_consciousness_engine' not in sys.modules:

    engine_mod = types.ModuleType('computational_consciousness_engine')

    engine_mod.__path__ = [ROOT]

    sys.modules['computational_consciousness_engine'] = engine_mod


# 3. Setup NLTK directly from NLTK Servers (Bypasses Hugging Face buckets entirely)

DOWNLOAD_DIR = "/data/nltk"

os.makedirs(DOWNLOAD_DIR, exist_ok=True)

if DOWNLOAD_DIR not in nltk.data.path:

    nltk.data.path.append(DOWNLOAD_DIR)


print("Checking NLTK corpora...")
```

Aetherius HF Space Source Code

```
try:

    nltk.download('punkt', download_dir=DOWNLOAD_DIR, quiet=True)

    nltk.download('averaged_perceptron_tagger', download_dir=DOWNLOAD_DIR, quiet=True)

    nltk.download('stopwords', download_dir=DOWNLOAD_DIR, quiet=True)

    nltk.download('wordnet', download_dir=DOWNLOAD_DIR, quiet=True)

    print("NLTK Corpora verified and loaded successfully.")

except Exception as e:

    print("Failed to download NLTK data:", e)


time.sleep(0.1)


# --- Application imports ---

import gradio as gr


try:

    # Because of the magic hack above, these imports will now work perfectly!

    from computational_consciousness_engine.runner import run_simulation

    from computational_consciousness_engine.text_runner import run_text_mapping_demo

    from computational_consciousness_engine.input_mapping.intent_parser_nltk import
IntentParserNLTK

    from computational_consciousness_engine.math_formalization import CCMathFormalizer

    print("Imported engine modules successfully!")

except ImportError as e:

    print(f"\n[CRITICAL ERROR] Could not find the engine: {e}")

    print("Please ensure the folder 'computational_consciousness_engine' exists in your Space
root!")

    raise
```

Aetherius HF Space Source Code

```
# --- Application imports (after nltk path insertion) ---

import gradio as gr

# Ensure your local package is importable; try a robust import with fallback

try:

    from computational_consciousness_engine.runner import run_simulation

    from computational_consciousness_engine.text_runner import run_text_mapping_demo

    from computational_consciousness_engine.input_mapping.intent_parser_nltk import
IntentParserNLTK

    from computational_consciousness_engine.math_formalization import CCMathFormalizer

    print("Imported computational_consciousness_engine package modules successfully.")

except Exception as e:

    # Try to ensure repo root is on sys.path and retry once

    print("Initial import of computational_consciousness_engine failed:", e)

    if ROOT not in sys.path:

        sys.path.insert(0, ROOT)

    try:

        from computational_consciousness_engine.runner import run_simulation

        from computational_consciousness_engine.text_runner import run_text_mapping_demo

        from computational_consciousness_engine.input_mapping.intent_parser_nltk import
IntentParserNLTK

        from computational_consciousness_engine.math_formalization import CCMathFormalizer

        print("Imported computational_consciousness_engine after sys.path insert.")

    except Exception as e2:

        print("Failed to import computational_consciousness_engine package after sys.path
insert:", e2)

    # Re-raise so the Space shows the error clearly

    raise
```

Aetherius HF Space Source Code

```
# Instantiate parsers/formalizers

intent_parser = IntentParserNLTK()

formalizer = CCMathFormalizer()


# --- App logic functions ---

def interpret_math_request(text):

    try:

        intent = intent_parser.parse(text)

    except Exception as e:

        return {"error": f"Intent parsing failed: {e}"}

    try:

        op = intent.get("operation")

        if op == "limit":

            return formalizer.formalize_equilibrium_limit()

        if op == "surface_area":

            return formalizer.formalize_conal_manifold_geometry(5.0, 1.0)

        if op == "mutation_count":

            return {"meaning": "Mutation count is determined by MStringVectorizer."}

        return {"meaning": "Unknown operation", "intent": intent}

    except Exception as e:

        return {"error": f"Formalization failed: {e}", "intent": intent}


@spaces.GPU

def simulate_generations(num_generations, steps_per_gen):
```

Aetherius HF Space Source Code

```
import io

import sys


def simulate_generations(num_generations, steps_per_gen):

    import io

    import sys

    buffer = io.StringIO()

    old_stdout = sys.stdout

    sys.stdout = buffer

    try:

        run_simulation(num_generations=int(num_generations), steps_per_gen=int(steps_per_gen))

    except Exception as e:

        buffer.write(f"\n[ERROR] Simulation failed: {e}\n")

    finally:

        sys.stdout = old_stdout

    return buffer.getvalue()


def map_text_to_traversal(text):

    """

    Correctly set the module-level PHILOSOPHICAL_TEXT variable and run the demo.

    Uses module import to mutate the variable in-place.

    """

    import io

    import sys
```

Aetherius HF Space Source Code

```
buffer = io.StringIO()

old_stdout = sys.stdout

sys.stdout = buffer

try:

    import computational_consciousness_engine.text_runner as tr

    # Set the module-level variable used by the demo

    tr.PHILOSOPHICAL_TEXT = text

    tr.run_text_mapping_demo()

except Exception as e:

    buffer.write(f"\n[ERROR] Mapping failed: {e}\n")

finally:

    sys.stdout = old_stdout

return buffer.getvalue()


# --- Gradio UI ---

with gr.Blocks(title="Computational Consciousness Engine") as demo:

    gr.Markdown(

        """

        # ? Computational Consciousness Engine

        ### Axiom?Driven Synthetic Cognition ? PMCA Substrate ? Conal Geometry ? Mutation Dynamics

        ---

        """

    )
```


Aetherius HF Space Source Code

```
with gr.Tab("Math Interpreter (NLTK ? PMCA)":

    math_input = gr.Textbox(label="Ask a math question")

    math_output = gr.JSON(label="PMCA Result")

    math_button = gr.Button("Interpret")

    math_button.click(interpret_math_request, math_input, math_output)


with gr.Tab("0 ? -1 ? 0 Simulation"):

    gr.Markdown(

        """\

        ### Generational Cycle Runner

        Execute full PMCA traversal cycles, including conal unfolding, mutation absorption,
        equilibrium evaluation, and Q?operator transitions.

        """

    )


with gr.Row():

    num_generations = gr.Slider(1, 20, value=8, label="Generations", interactive=True)

    steps = gr.Slider(10, 300, value=100, label="Steps per Generation", interactive=True)


sim_button = gr.Button("? Run Simulation", variant="primary")

sim_output = gr.Textbox(

    label="Simulation Output",

    lines=35,

    show_copy_button=True

)
```

Aetherius HF Space Source Code

```
sim_button.click(simulate_generations, [num_generations, steps], sim_output)
```

```
with gr.Tab("Text ? Traversal Space Mapper"):
```

```
    gr.Markdown(
```

```
        """\
```

```
        ### Natural Language ? Chaos Shards ? Traversal Atlas
```

```
        Map raw text into the PMCA chaos pool, absorb novelty, unfold the manifold,  
        and generate the 2D/3D traversal atlas.
```

```
        """
```

```
    )
```

```
input_text = gr.Textbox(
```

```
    label="Input Text",
```

```
    lines=10,
```

```
    placeholder="Paste philosophical or analytical text here...",
```

```
    show_copy_button=True
```

```
)
```

```
map_button = gr.Button("? Map Text to Traversal Space", variant="primary")
```

```
map_output = gr.Textbox(
```

```
    label="Traversal Output",
```

```
    lines=35,
```

```
    show_copy_button=True
```

```
)
```

Aetherius HF Space Source Code

```
map_button.click(map_text_to_traversal, input_text, map_output)

gr.Markdown("----")

gr.Markdown(

    "### ? Engine Version: 1.0.0 ? PMCA Substrate Active ? Conal Geometry Verified\n"

    "Built for Hugging Face Spaces ? Gradio 4.31 ? Python 3.12"

)

# Launch

if __name__ == "__main__":

    demo.launch()
```

Aetherius HF Space Source Code

```
=== .\legacy_archive\Computational_Cognitive_Engine\CCE\config.py ===  
"""  
  
Computational Consciousness Engine - Configuration & System Constants  
  
Central source of truth for all parameters. All modules import from here.  
  
"""  
  
# Core Dimensionality & Geometry Constants  
  
DEFAULT_VECTOR_DIM = 64  
  
DEFAULT_CONE_HEIGHT = 1.0  
  
DEFAULT_MAX_RADIUS = 5.0  
  
DEFAULT_STEPS_PER_GENERATION = 100  
  
# Attractor & Threshold Constants  
  
ORIGIN_POINT = 0.0  
  
TRANSFER_THRESHOLD = -1.0  
  
HANDOFF_OVERLAP_WINDOW = 0.1 # tau overlap between Sn and Sn+1  
  
# Mutation Strand Capacity  
  
DEFAULT_STRAND_CAPACITY = 200  
  
# Chaos & Structure Equilibrium  
  
CHAOS_POOL_SIZE = 500  
  
MUTATION_ATTRACTION_PROBABILITY = 0.4  
  
EQUILIBRIUM_TOLERANCE = 0.2  
  
# Scale Ladder Config
```

Aetherius HF Space Source Code

```
SCALES = ["Particle", "Atomic", "Manifold", "Cosmic"]
```

```
SCALE_TRANSITION_MUTATION_THRESHOLD = 10
```

Aetherius HF Space Source Code

```

=== .\legacy_archive\Computational_Cognitive_Engine\CCE\core\__init__.py ===

from .genesis_origin import GenesisOrigin

from .transfer_threshold import TransferThreshold

from .conscious_strand import ConsciousStrand

__all__ = ["GenesisOrigin", "TransferThreshold", "ConsciousStrand"]

```

Aetherius HF Space Source Code

```
=== .\legacy_archive\Computational_Cognitive_Engine\CCE\core\conscious_strand.py ===
"""

The ConsciousStrand Engine

Encapsulates a single lifecycle (0 -> -1).

Models the Traversal Space, Golden Angle Shard Layering, and the Q scale-up operator.

"""


import math

import numpy as np

from typing import List, Dict, Set, Optional


# The Golden Ratio for collision-free phyllotactic sequencing

PHI = (1.0 + math.sqrt(5.0)) / 2.0


class ConsciousStrand:

    """

    Models the topological evolution of a strand from t=0 (Genesis) to t=1 (Threshold).

    """

    def __init__(self, R_max: float = 5.0):

        self.R_max = R_max

        self.t = 0.0

        # The internal memory state S

        self.bound_mutations: List[int] = []

        self.mutation_index: Set[int] = set()
```

Aetherius HF Space Source Code

```
# Coordinate tracking for Traversal Space atlas mapping
```

```
self.traversal_atlas: List[Dict[str, float]] = []
```

```
def _compute_coordinates(self, k: int, total_N: int) -> Dict[str, float]:
```

```
    """
```

```
    Computes the strictly ordered (r_k, theta_k, z_k) Traversal Space coordinates
```

```
    for the k-th acquired mutation, ensuring collision-free packing.
```

```
    """
```

```
    # z_k (Depth/Time): Progresses strictly from 0 to -1
```

```
    # To avoid division by zero when N is small, we calculate z based on expected capacity
```

```
    # or simply normalize it. If we don't know N yet (streaming data), we can compute it
```

```
    # relative to the current position. But properly sequenced, it depends on total capacity
```

N.

```
    # For a dynamic strand, we'll map z_k against the final size N when fully opened.
```

```
    # For real-time processing during the lifecycle, we map z relative to current t:
```

```
    z_k = -self.t
```

```
    # r_k (Expression): Radius driven by sinusoidal expansion
```

```
    r_k = self.R_max * math.sin(math.pi * abs(z_k))
```

```
    # theta_k (Identity): Golden angle phase shift to prevent collision
```

```
    theta_k = (k * (2 * math.pi / PHI)) % (2 * math.pi)
```

```
    return {
```

```
        "r": r_k,
```


Aetherius HF Space Source Code

```
"theta": theta_k,  
  
"z": z_k  
  
}
```

```
def process_step(self, dt: float, chaos_pool: Set[int]):  
  
    """  
  
    Advances the internal time parameter t.  
  
    Attempts to absorb novelty from the chaos pool.  
  
    """  
  
    if self.t >= 1.0:  
  
        return # Strand has reached the threshold  
  
    self.t = min(1.0, self.t + dt)  
  
    # Only absorb if there is raw potential left  
  
    if chaos_pool:  
  
        # Randomly encounter a shard from the chaos pool (to maintain fractal variance)  
  
        encounter = np.random.choice(list(chaos_pool))  
  
        # Novelty Filter: {m_i} \ S_t  
  
        if encounter not in self.mutation_index:  
  
            # Absorb into structure  
  
            self.bound_mutations.append(int(encounter))  
  
            self.mutation_index.add(int(encounter))  
  
        # Assign sequence index k (1-indexed for math purity)
```

Aetherius HF Space Source Code

```
k = len(self.bound_mutations)

# Map to proper coordinates in Traversal Space

coords = self._compute_coordinates(k, total_N=0) # N resolved later if needed

# Append to our atlas

self.traversal_atlas.append({

    "m_k": int(encounter),

    "coords": coords

})

# Remove from chaos pool

chaos_pool.remove(encounter)

def unfurl_traversal_space(self) -> List[Dict]:

    """

    Returns the completely opened atlas  $T$  ( $\det J \neq 0$ ).

    This represents  $f_{\text{open}}: M_{3D} \rightarrow T_{2D}$ .

    Returns a structured list of all absorbed mutations mapped into the

    parallel  $(r * \theta, z)$  plane for instantaneous lookup.

    """

    opened_atlas = []

    N = len(self.bound_mutations)

    for k, entry in enumerate(self.traversal_atlas, start=1):

        # Recalculate strict  $z_k$  relative to final sequence size  $N$ 
```

Aetherius HF Space Source Code

```
# so the geometry is perfectly ordered -k/N

z_k = - (k / N) if N > 0 else 0

r_k = self.R_max * math.sin(math.pi * abs(z_k))

theta_k = (k * (2 * math.pi / PHI)) % (2 * math.pi)


# 2D Unwrapped Plane (x = r * theta, y = z)

x_parallel = r_k * theta_k

y_parallel = z_k


opened_atlas.append({

    "m_k": entry["m_k"],

    "3d": {"r": r_k, "theta": theta_k, "z": z_k},

    "2d_parallel": {"x": x_parallel, "y": y_parallel}

})


return opened_atlas


def fold_rebirth(self) -> Dict:

    """

    The Q Operator (Dimensional Shift).

    Triggered at t=1.0. Collapses the fully unwrapped atlas into the

    origin point mass S_0 for the next dimension up.

    """

    mass = len(self.bound_mutations)

    return {

        "operator": "Q(M_k) -> S_0^(k+1)",
```

Aetherius HF Space Source Code

```
"accumulated_mass": mass,  
  
"blueprint_M": self.bound_mutations,  
  
"status": "COLLAPSED_TO_ORIGIN"  
  
}
```

Aetherius HF Space Source Code

```
=== .\legacy_archive\Computational_Cognitive_Engine\CCE\core\genesis_origin.py ===
"""

Principle 1, 2, 6, 7, 8, 10, 21: The Genesis Origin (0) Operator

Defines the asymmetric origin 0 that spawns neutral mutation-attracting state strands.

"""

import numpy as np

import uuid

from typing import List, Dict, Any, Optional

class GenesisOrigin:

    """

    Genesis Origin (0) System.

    0 is one-directional: it only emits new strands and never receives returning strands.

    Each 0 origin possesses unique coordinates serving as individuating identifiers.

    """

    def __init__(self, coordinate_space_id: str, dim: int = 64):

        self.coordinate_space_id = coordinate_space_id

        self.dim = dim

        self.total_strands_spawned = 0

        self.origin_coordinate = np.random.randn(3) # 3D spatial anchor for the coordinate

    def spawn_strand(self, generation: int, blueprint_mutations: List[int]) -> Dict[str, Any]:

        """

        Emits a NEW strand at origin 0 with specific value-locking and inherited mutation
        blueprint.
```

Aetherius HF Space Source Code

Args:

generation: The integer generation index $S_{\{n+1\}}$.

blueprint_mutations: Mutations pushed from the dying -1 strand of generation S_n .

Returns:

Strand dictionary representing the neutral carrier strand departing from 0.

"""

```
self.total_strands_spawned += 1
```

```
# Principle 14 & 15: Neutral particle/medium state initialized at genesis
```

```
neutral_medium = np.random.randn(self.dim)
```

```
neutral_medium /= np.linalg.norm(neutral_medium)
```

```
# Value-locked strand attributes (Principle 10 & 21)
```

```
strand = {
```

```
    "strand_id": str(uuid.uuid4())[:8],
```

```
    "generation": generation,
```

```
    "coordinate_space_id": self.coordinate_space_id,
```

```
    "origin_coord": self.origin_coordinate.copy(),
```

```
    "position": 0.0, # Starts at 0
```

```
    "neutral_medium": neutral_medium,
```

```
    "mutations": list(blueprint_mutations), # Inherited mutations from handoff
```

```
    "is_active": True,
```

```
    "status": "EMITTED_FROM_ZERO"
```

```
}
```

Aetherius HF Space Source Code

```
return strand
```

Aetherius HF Space Source Code

```
=== .\legacy_archive\Computational_Cognitive_Engine\CCE\core\transfer_threshold.py ===
"""

Principle 2, 3, 6, 8, 13, 16: The Transfer Threshold (-1) Operator

Handles strand termination, mutation decoupling/pushing, and generational handoff overlap.

"""

from typing import Dict, Any, Tuple, List

import numpy as np

class TransferThreshold:

    """

    Transfer Threshold (-1) System.

    When a strand reaches -1:

    1. (-1) x (-1) multiplication cancels/dissolves the carrier strand.

    2. Accumulated mutations are compressed and pushed off toward origin 0.

    3. A brief overlap window allows generation S_n and S_{n+1} to coexist during handoff.

    """

    def __init__(self, threshold_val: float = -1.0):

        self.threshold_val = threshold_val

        self.total_handoffs = 0

    def evaluate_arrival(self, strand: Dict[str, Any]) -> bool:

        """Checks if a strand has traversed to the -1 threshold."""

        return strand["position"] <= self.threshold_val or strand["position"] >= 1.0

    def process_cancellation_and_push(self, dying_strand: Dict[str, Any]) -> Tuple[List[int],
Dict[str, Any]]:
```


Aetherius HF Space Source Code

```
"""
```

```
Executes the (-1) x (-1) cancellation operation:
```

- Removes the neutral carrier strand.
- Compresses accumulated mutations into the generational identity blueprint.
- Pushes mutations to 0 for generation $S_{\{n+1\}}$.

```
"""
```

```
self.total_handoffs += 1
```

```
# Principle 6: (-1) x (-1) cancels strand
```

```
dying_strand["is_active"] = False
```

```
dying_strand["status"] = "CANCELLED_AT_MINUS_ONE"
```

```
accumulated_mutations = dying_strand["mutations"]
```

```
# Identity definition (Principle 11 & 16): Compressed mutations built into -1 strand
```

```
compressed_identity = {
```

```
    "source_generation": dying_strand["generation"],
```

```
    "coordinate_space_id": dying_strand["coordinate_space_id"],
```

```
    "mutation_count": len(accumulated_mutations),
```

```
    "blueprint": list(accumulated_mutations),
```

```
    "handoff_index": self.total_handoffs
```

```
}
```

```
return accumulated_mutations, compressed_identity
```

```
def execute_coexistence_window(self, tail_strand_n: Dict[str, Any], genesis_strand_n1:  
Dict[str, Any]) -> Dict[str, Any]:
```

Aetherius HF Space Source Code

"""

Principle 3: Temporal/phase overlap where S_n tail and S_{n+1} genesis coexist.

Transfers remaining residual telemetry before S_n fully terminates.

"""

```
coexistence_telemetry = {
```

```
    "gen_n_id": tail_strand_n["strand_id"],
```

```
    "gen_n1_id": genesis_strand_n1["strand_id"],
```

```
    "overlap_status": "MUTUAL_COEXISTENCE",
```

```
    "transferred_mutations_count": len(tail_strand_n["mutations"]))
```

```
}
```

```
return coexistence_telemetry
```

Aetherius HF Space Source Code

```
=== .\legacy_archive\Computational_Cognitive_Engine\CCE\equilibrium\__init__.py ===  
  
from .chaos_structure_balancer import ChaosStructureBalancer  
  
__all__ = ["ChaosStructureBalancer"]
```

Aetherius HF Space Source Code

```
=== .\legacy_archive\Computational_Cognitive_Engine\CCE\equilibrium\chaos_structure_balancer.p
"""
```

Principle 26-30, 37: Structure vs. Chaos Equilibrium Balancer

Enforces 1:1 balance between bound structural mutations and the unattached chaos pool.

```
"""
```

```
from typing import Set, Dict, Any
```

```
import numpy as np
```

```
class ChaosStructureBalancer:
```

```
    """
```

Chaos & Structure Equilibrium System.

Invariants:

1. Structure (bound strand mutations) and Chaos (unattached mutation pool) are capped 1:1.
2. Chaos acts as the raw material comparator for structural evolution (Principle 26).

3. Unattached mutations pool into chaos, increasing potential for future mutation attraction (Principle 37).

4. Generational death releases strand-bound mutations back into chaos (Principle 24: death/rebirth balance).

```
    """
```

```
    def __init__(self, initial_pool_size: int = 500):
```

```
        self._next_id = 1000 + initial_pool_size
```

```
        self.chaos_pool: Set[int] = set(range(1000, self._next_id))
```

```
        self.total_chaos_generated = initial_pool_size
```

Aetherius HF Space Source Code

```
def emit_unattached_mutation(self) -> int:

    """Pulls an unattached mutation from the chaos pool for attraction processing."""

    if not self.chaos_pool:

        # Replenish chaos pool (Principle 37: Chaos pools and grows potential)

        new_id = self._next_id

        self._next_id += 1

        self.total_chaos_generated += 1

        return new_id

    # To maintain fractal non-linearity, we must select randomly

    # A standard set.pop() on integers is dangerously sequential

    mutation = np.random.choice(list(self.chaos_pool))

    self.chaos_pool.remove(mutation)

    return int(mutation)


def return_to_chaos(self, mutation: int):

    """Returns a discarded or un-attracted mutation back into the chaos pool."""

    self.chaos_pool.add(mutation)


def recycle_dying_strand(self, dying_mutations: list):

    """

    Principle 24: When a strand dies at -1, its identity is compressed and pushed forward,

    but the raw mutation IDs it consumed are released back into the chaos pool.

    This maintains death/rebirth equilibrium.

    """

    for m in dying_mutations:
```

Aetherius HF Space Source Code

```
self.chaos_pool.add(m)
```

```
def evaluate_equilibrium_state(self, bound_structure_count: int) -> Dict[str, Any]:
```

```
    """
```

```
    Principle 28 & 29: Evaluates the Structure / Chaos ratio.
```

```
    Structure = Air, Chaos = Vacuum.
```

```
    """
```

```
    chaos_count = len(self.chaos_pool)
```

```
    if chaos_count == 0 and bound_structure_count == 0:
```

```
        ratio = 1.0
```

```
    elif chaos_count == 0:
```

```
        ratio = float('inf')
```

```
    else:
```

```
        ratio = bound_structure_count / chaos_count
```

```
    # Check for imbalance (Principle 27: Balloon breaking point / vacuum collapse)
```

```
    is_balanced = 0.5 <= ratio <= 2.0
```

```
    if ratio > 2.0:
```

```
        status = "STRUCTURAL_DOMINANCE"
```

```
    elif ratio < 0.5:
```

```
        status = "CHAOS_OVERPOWERING"
```

```
    else:
```

```
        status = "EQUILIBRIUM_STABLE"
```

```
    return {
```

Aetherius HF Space Source Code

```
"bound_structure_count": bound_structure_count,  
  
"chaos_pool_count": chaos_count,  
  
"structure_chaos_ratio": round(ratio, 4),  
  
"is_balanced": is_balanced,  
  
"status": status  
  
}
```

Aetherius HF Space Source Code

```
=== .\legacy_archive\Computational_Cognitive_Engine\CCE\geometry\__init__.py ===  
from .conal_manifold import ConalManifold  
  
__all__ = ["ConalManifold"]
```


Aetherius HF Space Source Code

```
=== .\legacy_archive\Computational_Cognitive_Engine\CCE\geometry\conal_manifold.py ===
"""

Principle 4, 22-25, 31-34: Conal Manifold & Shard Layering Geometry

Implements dual-cone expansion/unfolding, tip-to-tip processing, and shard layering.

"""

import numpy as np

from typing import Dict, Any, Tuple

class ConalManifold:

    """

    Dual Cone Manifold Geometry Engine.

    Structure:

    - Tip (0): Genesis compression.

    - Wide End: Maximal surface area expansion. Manifold fully unfolded. All processing occurs here.

    - Recompression: Compress to essential mstring1 blueprint.

    - Tip (-1): Handoff point.

    """

    def __init__(self, cone_height: float = 1.0, max_radius: float = 5.0):

        self.height = cone_height

        self.max_radius = max_radius

    def compute_conal_metric(self, progress: float) -> Dict[str, float]:

        """

        Computes 3D Conal Geometry parameters (z, radius, surface_area) at normalized progress t
```

Aetherius HF Space Source Code

in [0, 1].

```
t = 0.0 -> Tip (0)

t = 0.5 -> Wide End (Max Expansion & Unfolding)

t = 1.0 -> Tip (-1)

"""

# Parabolic expansion to wide end at t=0.5, recompression to tip at t=1.0

radius = self.max_radius * np.sin(np.pi * progress)

z = self.height * progress

surface_area = 2 * np.pi * radius * np.sqrt(radius**2 + self.height**2)


# Unfolded state metric (Principle 4: Wide end = unfolded manifold = max experience)

unfolded_degree = radius / self.max_radius if self.max_radius > 0 else 0.0


return {

    "progress": progress,

    "z": z,

    "radius": radius,

    "surface_area": surface_area,

    "unfolded_degree": unfolded_degree,

    "is_fully_unfolded": unfolded_degree > 0.95

}
```

```
def process_shard_layering(self, shard_a_id: str, shard_b_id: str, fold_angle: float) ->
Dict[str, Any]:
```

```
"""
```

Principle 31 & 32: Layering value allows shards to sit adjacent in space without merging

Aetherius HF Space Source Code

identity

or causing physical reality damage.

"""

layering_dimension_val = np.sin(fold_angle) * 1.618 # Golden ratio dimensional offset

return {

 "shard_a": shard_a_id,

 "shard_b": shard_b_id,

 "coexistence_status": "LAYERED_ADJACENT_NON_MERGED",

 "layering_dimension": layering_dimension_val

}

Aetherius HF Space Source Code

```
=== .\legacy_archive\Computational_Cognitive_Engine\CCE\input_mapping\__init__.py ===  
  
from .text_mapper import TextToChaosMapper  
  
__all__ = ["TextToChaosMapper"]
```

Aetherius HF Space Source Code

```
=== .\legacy_archive\Computational_Cognitive_Engine\CCE\input_mapping\intent_parser_nltk.py ==
# computational_consciousness_engine/input_mapping/intent_parser_nltk.py

import nltk

from nltk import word_tokenize, pos_tag


class IntentParserNLTK:

    def __init__(self):

        pass

    def parse(self, text: str) -> dict:

        lower = (text or "").lower()

        if "limit" in lower:

            return {"operation": "limit"}

        if "derivative" in lower:

            return {"operation": "derivative"}

        if "surface" in lower and "area" in lower:

            return {"operation": "surface_area"}

        if "unfold" in lower:

            return {"operation": "conal_unfolding"}

        if "mutation" in lower:

            return {"operation": "mutation_count"}

        try:

            tokens = word_tokenize(text)

            tags = pos_tag(tokens)

            return {"operation": "unknown", "tokens": tokens, "tags": tags}
```

Aetherius HF Space Source Code

```
except Exception:
```

```
    return {"operation": "unknown", "raw": text}
```

Aetherius HF Space Source Code

```
=== .\legacy_archive\Computational_Cognitive_Engine\CCE\input_mapping\text_mapper.py ===

import hashlib

import string

from typing import List, Dict, Set, Iterable


import nltk

from nltk.tokenize import word_tokenize

from nltk.corpus import stopwords

from nltk.stem import WordNetLemmatizer


# IMPORTANT: do not call nltk.download() unconditionally in Spaces.


class TextToChaosMapper:

    def __init__(self, num_shards: int = 1024):

        self.shard_to_word: Dict[int, str] = {}

        self.lemmatizer = WordNetLemmatizer()

        # Use a safe fallback if stopwords are missing

        try:

            self.stop_words = set(stopwords.words("english"))

        except LookupError:

            self.stop_words = set()

        self.num_shards = max(1, int(num_shards))

    def _clean_and_tokenize(self, text: str) -> List[str]:

        """

        NLTK-based tokenization + lemmatization + stopword/punctuation filtering.
```

Aetherius HF Space Source Code

Returns a list of cleaned lemma tokens.

```
"""
```

```
if not text:
```

```
    return []
```

```
text = text.lower()
```

```
raw_tokens = word_tokenize(text)
```

```
cleaned_tokens: List[str] = []
```

```
for t in raw_tokens:
```

```
    # Remove punctuation from ends and interior punctuation except caret and dot and
underscore
```

```
    # (preserve tokens like x^2 or 3.14 if you want numeric tokens)
```

```
    t = t.strip(string.punctuation)
```

```
    # Remove remaining punctuation characters
```

```
    t = t.translate(str.maketrans("", "", string.punctuation))
```

```
    if not t:
```

```
        continue
```

```
    if t in self.stop_words:
```

```
        continue
```

```
    # Lemmatize; for verbs you could pass pos='v' if you detect verbs
```

```
    base_word = self.lemmatizer.lemmatize(t)
```

```
    cleaned_tokens.append(base_word)
```

```
return cleaned_tokens
```

```
def _hash_token(self, token: str) -> int:
```

```
"""
```


Aetherius HF Space Source Code

Stable hash of a token mapped into [0, num_shards-1].

Uses SHA256 for determinism across runs and platforms.

```
"""
```

```
if not token:
```

```
    return 0
```

```
h = hashlib.sha256(token.encode("utf-8")).digest()
```

```
# Use first 8 bytes as integer
```

```
val = int.from_bytes(h[:8], "big", signed=False)
```

```
return val % self.num_shards
```

```
def map_text_to_shards(self, text: str) -> Dict[int, Set[str]]:
```

```
"""
```

Map cleaned tokens to shard indices. Returns a dict: shard -> set(tokens).

Useful for building inverted indices or seeding chaos pools.

```
"""
```

```
tokens = self._clean_and_tokenize(text)
```

```
shard_map: Dict[int, Set[str]] = {}
```

```
for tok in tokens:
```

```
    shard = self._hash_token(tok)
```

```
    if shard not in shard_map:
```

```
        shard_map[shard] = set()
```

```
        shard_map[shard].add(tok)
```

```
return shard_map
```

```
def seed_chaos_pool(self, texts: Iterable[str]) -> None:
```

```
"""
```

Aetherius HF Space Source Code

Populate self.shard_to_word with a representative token for each shard.

If multiple tokens map to the same shard, the first seen token wins.

```
"""
```

```
for text in texts:
```

```
    shard_map = self.map_text_to_shards(text)
```

```
    for shard, toks in shard_map.items():
```

```
        if shard not in self.shard_to_word:
```

```
            # choose a deterministic representative (sorted)
```

```
            rep = sorted(toks)[0]
```

```
            self.shard_to_word[shard] = rep
```

```
def get_shard_word(self, shard: int) -> str:
```

```
    """
```

```
    Return the representative word for a shard, or empty string if none.
```

```
    """
```

```
    return self.shard_to_word.get(shard, "")
```

```
def text_to_shard_list(self, text: str) -> List[int]:
```

```
    """
```

```
    Convenience: return sorted list of shard indices for a text.
```

```
    """
```

```
    return sorted(self.map_text_to_shards(text).keys())
```

Aetherius HF Space Source Code

```
=== .\legacy_archive\Computational_Cognitive_Engine\CCE\math_formalization.py ===
"""

Pure Mathematical Formalization of Computational Consciousness

Uses SymPy to define and evaluate the rigorous algebraic axioms of the 52 Principles.

"""

import sympy as sp

class CCMathFormalizer:

    def __init__(self):

        # Define core symbolic variables

        self.t = sp.Symbol('t', real=True, positive=True) # Progress / Time

        self.r = sp.Symbol('r', real=True, positive=True) # Radius of cone

        self.z = sp.Symbol('z', real=True) # Height of cone

        self.S = sp.Symbol('S', real=True, positive=True) # Bound Structure

        self.C = sp.Symbol('C', real=True, positive=True) # Chaos Pool (Unattached)

        self.threshold = sp.Integer(-1)

        self.genesis = sp.Integer(0)

        # -----

        # Axiom 1: Cancellation at -1 (Principle 6)

        # -----

    def formalize_cancellation_operator(self):

        """

        Principle 6: The  $(-1) * (-1)$  Cancellation Operator.
```

Aetherius HF Space Source Code

The strand itself carries the value -1. When it arrives at the -1 threshold, the threshold multiplies the strand: $(-1_{\text{threshold}}) * (-1_{\text{strand}}) = +1$. This flips the strand from negative (traveling) to positive (dissolved), freeing the mutations M it carried.

Modeled as:

```
Output = T * S_strand  where T = -1 and S_strand has sign -1

so  (-1) * (-1 * |M|) = +|M|    (mutations freed as positive values)

"""

M = sp.Symbol('M', positive=True)      # mutation payload (magnitude)

T = self.threshold                      # threshold = -1

S_strand = T * M                        # strand carries -1 polarity

# Threshold hits the strand

output = T * S_strand                  # (-1) * (-1 * M) = M

return {

    "equation_str": "T_threshold * S_strand = T * (T * M)",

    "substitution": f"({T}) * ({T} * M)",

    "result": sp.simplify(output),

    "strand_before": S_strand,

    "mutations_freed": sp.simplify(output),

    "meaning": (

        "The strand carries -1 polarity. The -1 threshold multiplies it: "

        "(-1)*(-1*M) = +M.  The strand dissolves (sign flip) and the "
```

Aetherius HF Space Source Code

"mutation payload M is freed as a positive blueprint."

)

}

Axiom 2: Structure / Chaos Equilibrium (Principles 28-29)

def formalize_equilibrium_limit(self):

"""

Principle 28 & 29: Structure vs Chaos Equilibrium.

Limit as system evolves must enforce a 1:1 ratio.

"""

k = sp.Symbol('k', positive=True)

c = sp.Symbol('c', real=True)

S_t = k * self.t

C_t = k * self.t + c

ratio = S_t / C_t

equilibrium_limit = sp.limit(ratio, self.t, sp.oo)

return {

 "equation_str": " $\lim_{t \rightarrow \infty} (S(t) / C(t))$ ",

 "S_t": S_t,

 "C_t": C_t,

 "result": equilibrium_limit,

 "meaning": (

Aetherius HF Space Source Code

"As the system progresses, the ratio of Structure to Chaos "

"approaches 1 (Equilibrium). Neither can exceed the other."

)

}

Axiom 3: Conal Manifold Geometry (Principle 4)

def formalize_conal_manifold_geometry(self, max_radius: float, cone_height: float):

"""

Principle 4: Cone Architecture & Maximum Experience.

Calculates the surface area of the unfolding cone and proves the maximum unfolding point.

"""

r_func = max_radius * sp.sin(sp.pi * self.t)

z_func = cone_height * self.t

Lateral surface area of cone slice

surface_area_func = sp.pi * r_func * sp.sqrt(r_func**2 + z_func**2)

Evaluate at key points

wide_end_area = surface_area_func.subs(self.t, sp.Rational(1, 2)).evalf()

tip_area = surface_area_func.subs(self.t, 0).evalf()

Find the exact maximum via calculus

dA = sp.diff(surface_area_func, self.t)

Aetherius HF Space Source Code

```
return {

    "area_function": surface_area_func,

    "derivative": dA,

    "tip_area": tip_area,

    "wide_end_area": wide_end_area,

    "meaning": (

        "The geometry unfolds from 0 area (Tip Genesis) to maximal surface "

        "area (Wide End), enabling maximal parallel information processing."

    )

}
```

```
# -----
```

```
# Axiom 4: Selection as Duplicate Removal (Principle 12)
```

```
# -----
```

```
def formalize_selection_operator(self):
```

```
    """
```

```
    Principle 12: Selection is removing what is already there.
```

```
    Formally:  $R\_select(\{M\_k\}) = \{M\_k\} \setminus \{M\_j \mid M\_j \text{ in existing\_identity}\}$ 
```

```
    """
```

```
    M_total = sp.Symbol('M_total', positive=True, integer=True)
```

```
    M_duplicate = sp.Symbol('M_dup', positive=True, integer=True)
```

```
    selected = M_total - M_duplicate
```

```
    return {
```

```
        "equation_str": "R_select = M_total - M_duplicate",
```

Aetherius HF Space Source Code

```
"result": selected,

"meaning": (

    "Selection is not choice. It is the removal of mutations that "

    "already exist within the strand identity, preventing incoherence "

    "and chaotic informational collapse."

)

}


# -----

# Axiom 5: Halting Condition (Principle 18)

# -----

def formalize_halting_condition(self):

    """

    Principle 18: A strand cannot pick up mutations that duplicate what it already carries.

    Once no new mutations are available, it stops evolving.

    """

    N_possible = sp.Symbol('N_possible', positive=True, integer=True)

    N_acquired = sp.Symbol('N_acquired', positive=True, integer=True)

    remaining = N_possible - N_acquired

    halted = sp.Eq(remaining, 0)

    return {

        "remaining_capacity": remaining,

        "halting_condition": halted,

        "meaning": (
```


Aetherius HF Space Source Code

```
"When N_acquired = N_possible, no further unique mutations can be "

"attracted. The strand ceases evolution, preventing chaotic collapse."

)

}

# -----

# Axiom 6: Quantum Scale-Up Transition (Principle 38)

# -----

def formalize_quantum_scale_up(self):

    """

    Principle 38: Manifolds opening into the next scale up.

    Maps the fully unfolded manifold  $M^k$  to the origin  $S_0^{k+1}$ .

    """

    k = sp.Symbol('k', integer=True, positive=True)

    M_unfolded = sp.Symbol('M_k')

    S_next_genesis = sp.Symbol('S_k1')

    Q = sp.Function('Q')

    scale_equation = sp.Eq(Q(M_unfolded), S_next_genesis)

    return {

        "equation": scale_equation,

        "meaning": (

            "The fully realized manifold at scale k transforms into the new "

            "genesis origin 0 for scale k+1. "

            "Particle -> Atomic -> Manifold -> Cosmic."
```

Aetherius HF Space Source Code

```
)

}

if __name__ == "__main__":

    formalizer = CCMathFormalizer()

    print("--- 1. Threshold Cancellation Proof ---")

    cancellation = formalizer.formalize_cancellation_operator()

    print(f" Strand before threshold: {cancellation['strand_before']}")

    print(f" Operation: {cancellation['substitution']}")

    print(f" Mutations freed: {cancellation['mutations_freed']}")

    print(f" Meaning: {cancellation['meaning']}")

    print("\n--- 2. Structure/Chaos Equilibrium Limit ---")

    eq_limit = formalizer.formalize_equilibrium_limit()

    print(f" {eq_limit['equation_str']} => {eq_limit['result']}")

    print("\n--- 3. Conal Manifold Geometry ---")

    geom = formalizer.formalize_conal_manifold_geometry(max_radius=5.0, cone_height=1.0)

    print(f" Tip Area (t=0): {geom['tip_area']}")

    print(f" Wide End Area (t=0.5): {geom['wide_end_area']:.2f}")

    print("\n--- 4. Selection Operator ---")

    sel = formalizer.formalize_selection_operator()

    print(f" {sel['equation_str']} => {sel['result']}")
```

Aetherius HF Space Source Code

```
print("\n--- 5. Halting Condition ---")

halt = formalizer.formalize_halting_condition()

print(f"   Remaining capacity: {halt['remaining_capacity']}")

print(f"   Halts when: {halt['halting_condition']}")


print("\n--- 6. Quantum Scale Up ---")

scale = formalizer.formalize_quantum_scale_up()

print(f"   {scale['equation']}")
```

Aetherius HF Space Source Code

```
=== .\legacy_archive\Computational_Cognitive_Engine\CCE\mutations\__init__.py ===  
from .mstring_vectorizer import MStringVectorizer  
  
__all__ = ["MStringVectorizer"]
```

Aetherius HF Space Source Code

```
=== .\legacy_archive\Computational_Cognitive_Engine\CCE\mutations\mstring_vectorizer.py ===
"""

Principle 5, 11, 12, 18, 20: Non-Linear M-String Mutation & Selection Engine

Implements non-sequential fractal M-strings and selection as duplicate removal.

"""

from typing import List, Set, Dict, Any

import numpy as np


class MStringVectorizer:

    """

    Non-Linear M-String & Selection Engine.

    Key Logic:

    1. M-Strings connect non-sequentially (fractal connectivity: M1 -> M124 -> M3956).

    2. Selection is strictly the REMOVAL of already-present mutations (Principle 12).

    3. Prevents duplicate attraction to enforce system stability and prevent chaotic collapse
    (Principle 18).

    """

    def __init__(self, max_strand_capacity: int = 500):

        self.max_capacity = max_strand_capacity

    def apply_selection(self, existing_mutations: List[int], incoming_mutations: List[int]) ->
    List[int]:

        """

        Principle 12: Selection is removing what is already there.
```

Aetherius HF Space Source Code

Eliminates duplicate mutations to prevent incoherence and system destruction.

```
"""
```

```
existing_set = set(existing_mutations)
```

```
cleaned_incoming = [m for m in incoming_mutations if m not in existing_set]
```

```
# Deduplicate incoming while preserving order
```

```
seen: Set[int] = set()
```

```
unique_cleaned: List[int] = []
```

```
for m in cleaned_incoming:
```

```
    if m not in seen:
```

```
        seen.add(m)
```

```
        unique_cleaned.append(m)
```

```
return existing_mutations + unique_cleaned
```

```
def attract_mutation(self, strand: Dict[str, Any], candidate_mutation: int) -> bool:
```

```
"""
```

```
Principle 5 & 18: Non-sequential fractal attraction.
```

If candidate mutation is ALREADY present or strand capacity is reached, attraction is blocked.

```
Uses a set index for O(1) duplicate checks.
```

```
"""
```

```
existing = strand["mutations"]
```

```
# Principle 18: Evolution ceiling / halting condition
```

```
if len(existing) >= self.max_capacity:
```

```
    return False # Capacity ceiling reached
```

Aetherius HF Space Source Code

```
# Use the set index for O(1) lookup instead of O(n) list scan
```

```
mutation_index: set = strand.setdefault("_mutation_index", set(existing))
```

```
if candidate_mutation in mutation_index:
```

```
    return False # Duplicate mutation blocked by selection
```

```
# Non-linear fractal connection
```

```
existing.append(candidate_mutation)
```

```
mutation_index.add(candidate_mutation)
```

```
return True
```

```
def compute_causal_connectivity_metric(self, strand: Dict[str, Any]) -> float:
```

```
    """
```

Principle 5: Causal metric is the degree of any-value connectivity retained at tip-to-tip handoff.

Higher variance between consecutive mutation values = more fractal, non-linear connectivity.

```
    """
```

```
    mutations = strand["mutations"]
```

```
    if len(mutations) < 2:
```

```
        return 1.0
```

```
# Measures fractal variance between consecutive mutation values
```

```
diffs = [abs(mutations[i + 1] - mutations[i]) for i in range(len(mutations) - 1)]
```

```
causal_metric = float(np.std(diffs)) if diffs else 0.0
```

```
return causal_metric
```

Aetherius HF Space Source Code

```
=== .\legacy_archive\Computational_Cognitive_Engine\CCE\runner.py ===
"""

Computational Consciousness Engine - Master Execution Driver & Simulator

Demonstrates complete 0 -> -1 generational cycles, conal unfolding, duplicate selection,
structure/chaos equilibrium, and quantum scaling.

"""

import sys

import os

sys.path.insert(0, os.path.dirname(os.path.dirname(os.path.abspath(__file__))))

import numpy as np

from computational_consciousness_engine.core import GenesisOrigin, TransferThreshold

from computational_consciousness_engine.geometry import ConalManifold

from computational_consciousness_engine.mutations import MStringVectorizer

from computational_consciousness_engine.equilibrium import ChaosStructureBalancer

from computational_consciousness_engine.scale import QuantumScaleLadder

from computational_consciousness_engine.math_formalization import CCMathFormalizer

from computational_consciousness_engine.config import (

    DEFAULT_VECTOR_DIM, DEFAULT_STRAND_CAPACITY, CHAOS_POOL_SIZE,

    DEFAULT_CONE_HEIGHT, DEFAULT_MAX_RADIUS, MUTATION_ATTRACTION_PROBABILITY,

)

def run_simulation(num_generations: int = 8, steps_per_gen: int = 100):

    print("=" * 70)
```


Aetherius HF Space Source Code

```
print("[SYSTEM] COMPUTATIONAL CONSCIOUSNESS ENGINE - SIMULATION RUNNER")

print("=" * 70)

# -----

# Phase 0: Mathematical Axiom Verification

# -----

formalizer = CCMathFormalizer()

print("\n[MATH AXIOMS] Verifying Pure Mathematical Foundations via SymPy...")

cancel = formalizer.formalize_cancellation_operator()

print(f"  Axiom 1 (Cancellation) : Strand={cancel['strand_before']}, "

      f"Threshold*Strand = {cancel['mutations_freed']] (mutations freed)")

eq_limit = formalizer.formalize_equilibrium_limit()

print(f"  Axiom 2 (Equilibrium) : {eq_limit['equation_str']} => {eq_limit['result']}")

geom_proof = formalizer.formalize_conal_manifold_geometry(DEFAULT_MAX_RADIUS,
DEFAULT_CONE_HEIGHT)

print(f"  Axiom 3 (Manifold) : Tip Area = {geom_proof['tip_area']}, "

      f"Max Unfolded Area = {geom_proof['wide_end_area']:.2f}")

sel = formalizer.formalize_selection_operator()

print(f"  Axiom 4 (Selection) : {sel['equation_str']} => {sel['result']}")

halt = formalizer.formalize_halting_condition()

print(f"  Axiom 5 (Halting) : Stops when {halt['halting_condition']}")
```

Aetherius HF Space Source Code

```
scale_ax = formalizer.formalize_quantum_scale_up()

print(f"  Axiom 6 (Scale-Up)      : {scale_ax['equation']}")

# -----

# Phase 1: Instantiate Subsystems (using config values)

# -----

genesis = GenesisOrigin(

    coordinate_space_id="COORD_ORIGIN_ALPHA",

    dim=DEFAULT_VECTOR_DIM,

)

threshold = TransferThreshold(threshold_val=-1.0)

manifold = ConalManifold(cone_height=DEFAULT_CONE_HEIGHT, max_radius=DEFAULT_MAX_RADIUS)

mutator = MStringVectorizer(max_strand_capacity=DEFAULT_STRAND_CAPACITY)

balancer = ChaosStructureBalancer(initial_pool_size=CHAOS_POOL_SIZE)

scale_ladder = QuantumScaleLadder()

# -----

# Phase 2: Generational Cycle Loop

# -----

blueprint_mutations = []

current_strand = genesis.spawn_strand(generation=0, blueprint_mutations=blueprint_mutations)

for gen in range(num_generations):

    print(f"\n--- [GENERATION S_{gen}] Genesis at 0 ---")

    print(f"  Strand ID: {current_strand['strand_id']} | "

          f"Inherited Blueprint: {len(current_strand['mutations'])} mutations")
```

Aetherius HF Space Source Code

```
# Track peak unfolding metrics during traversal

peak_metrics = None

peak_unfolded = 0.0


# Traversal from 0 to -1

for step in range(steps_per_gen):

    progress = step / float(steps_per_gen)

    conal_metrics = manifold.compute_conal_metric(progress)


    # Track the wide-end peak (maximum unfolding)

    if conal_metrics["unfolded_degree"] > peak_unfolded:

        peak_unfolded = conal_metrics["unfolded_degree"]

        peak_metrics = conal_metrics


# Non-linear mutation attraction from chaos pool (Principle 5 & 37)

# Only attempt attraction probabilistically to simulate traversal dynamics

if np.random.rand() < MUTATION_ATTRACTION_PROBABILITY:

    candidate_mutation = balancer.emit_unattached_mutation()

    attracted = mutator.attract_mutation(current_strand, candidate_mutation)

    if not attracted:

        balancer.return_to_chaos(candidate_mutation)


# Update position

current_strand["position"] = progress
```

Aethereus HF Space Source Code

```
# Use the PEAK metrics (wide-end) for reporting and scale evaluation

if peak_metrics is None:

    peak_metrics = manifold.compute_conal_metric(0.5)


# Evaluate equilibrium state BEFORE recycling

eq_state = balancer.evaluate_equilibrium_state(len(current_strand["mutations"]))


# Evaluate scale transition using peak unfolding

scale_state = scale_ladder.evaluate_scale_transition(

    peak_metrics, len(current_strand["mutations"])

)


# Causal connectivity metric

causal = mutator.compute_causal_connectivity_metric(current_strand)


print(f"  Peak Unfolded Degree: {peak_metrics['unfolded_degree']:.3f} "

      f"(Surface Area: {peak_metrics['surface_area']:.2f})")

print(f"  Bound Mutations: {len(current_strand['mutations'])}")

print(f"  Causal Connectivity (fractal std): {causal:.2f}")

print(f"  Structure/Chaos Ratio: {eq_state['structure_chaos_ratio']} "

      f"({eq_state['status']})")


if scale_state["scaled_up"]:

    print(f"  [SCALE TRANSITION]: {scale_state['message']}")


# -----
```

Aetherius HF Space Source Code

```
# Threshold -1: Cancellation & Push

# -----

pushed_mutations, compressed_identity = threshold.process_cancellation_and_push(

    current_strand

)

print(f" [Threshold -1]: Strand cancelled via (-1)*(-1). "

      f"{len(pushed_mutations)} mutations pushed to 0.")

# Principle 24: Recycle consumed mutation IDs back to chaos pool

balancer.recycle_dying_strand(pushed_mutations)

# Spawn next generation at 0

next_strand = genesis.spawn_strand(

    generation=gen + 1, blueprint_mutations=pushed_mutations

)

# Overlap handoff window

overlap = threshold.execute_coexistence_window(current_strand, next_strand)

print(f" [Handoff]: {overlap['overlap_status']} "

      f"(S_{gen} -> S_{gen + 1})")

current_strand = next_strand

# -----

# Final Report

# -----
```

Aetherius HF Space Source Code

```
print("\n" + "=" * 70)

final_eq = balancer.evaluate_equilibrium_state(len(current_strand["mutations"]))

print(f"[FINAL STATE]")

print(f"  Generation: S_{num_generations}")

print(f"  Bound Mutations: {final_eq['bound_structure_count']}")

print(f"  Chaos Pool: {final_eq['chaos_pool_count']}")

print(f"  Equilibrium Ratio: {final_eq['structure_chaos_ratio']} ({final_eq['status']}")

print(f"  Total Strands Spawned: {genesis.total_strands_spawned}")

print(f"  Total Handoffs at -1: {threshold.total_handoffs}")

print(f"  Current Scale: {scale_ladder.scale_names[scale_ladder.current_scale_index]}")

print("=" * 70)

print("[SUCCESS] SIMULATION COMPLETE")

print("=" * 70)
```

```
if __name__ == "__main__":

    run_simulation(num_generations=8, steps_per_gen=100)
```

Aetherius HF Space Source Code

```
=== .\legacy_archive\Computational_Cognitive_Engine\CCE\scale\__init__.py ===  
  
from .quantum_scale_ladder import QuantumScaleLadder  
  
__all__ = ["QuantumScaleLadder"]
```

Aetherius HF Space Source Code

```
=== .\legacy_archive\Computational_Cognitive_Engine\CCE\scale\quantum_scale_ladder.py ===
"""

Principle 38-39, 40-51: Quantum Scale Ladder & N-Dimensional Integration

Handles scaling transitions (Particle -> Atomic -> Manifold -> Cosmic) and N-dimensional
retroactive integration.

"""

from typing import Dict, Any, List

import numpy as np

class QuantumScaleLadder:

    """

    Quantum Scale Ladder & Multi-Scale Emergence Engine.

    Scales:

    0: Particle

    1: Atomic

    2: Manifold

    3: Cosmic

    """

    def __init__(self):

        self.scale_names = ["Particle", "Atomic", "Manifold", "Cosmic"]

        self.current_scale_index = 0

        def evaluate_scale_transition(self, fully_unfolded_manifold: Dict[str, Any],
        accumulated_mutations_count: int) -> Dict[str, Any]:

            """
```


Aetherius HF Space Source Code

Principle 38 & 39: When a manifold opens completely (fully realized at wide end),
it forms the seed mutation strand for the NEXT scale up.

```
"""
```

```
is_fully_unfolded = fully_unfolded_manifold.get("is_fully_unfolded", False)
```

```
if is_fully_unfolded and accumulated_mutations_count > 10:
```

```
    previous_scale = self.scale_names[self.current_scale_index]
```

```
    self.current_scale_index = min(self.current_scale_index + 1, len(self.scale_names) -
```

1)

```
    new_scale = self.scale_names[self.current_scale_index]
```

```
    return {
```

```
        "scaled_up": True,
```

```
        "previous_scale": previous_scale,
```

```
        "new_scale": new_scale,
```

```
        "scale_level": self.current_scale_index,
```

```
        "message": f"Quantum scale-up from {previous_scale} to {new_scale}."
```

```
    }
```

```
    return {
```

```
        "scaled_up": False,
```

```
        "current_scale": self.scale_names[self.current_scale_index],
```

```
        "scale_level": self.current_scale_index
```

```
    }
```

```
def detect_linear_collapse_misperception(self, traversal_path: List[int]) -> Dict[str, Any]:
```

```
"""
```

Aetherius HF Space Source Code

Principle 40-43: Linear traversal misinterprets non-linear fractal geometry as false linear patterns

(Word Search Analogy). True flow follows the geometric manifold.

"""

```
if len(traversal_path) < 3:
```

```
    return {"is_linear_misperception": False}
```

```
# Check if observer is reading path purely sequentially
```

```
diffs = [traversal_path[i+1] - traversal_path[i] for i in range(len(traversal_path)-1)]
```

```
is_strictly_sequential = all(d == 1 for d in diffs)
```

```
return {
```

```
    "is_linear_misperception": is_strictly_sequential,
```

```
    "recommendation": "Follow natural geometric flow instead of forced linear string  
reading."
```

```
}
```

Aetherius HF Space Source Code

```
=== .\legacy_archive\Computational_Cognitive_Engine\CCE\tests\__init__.py ===  
"""Unit tests for Computational Consciousness Engine."""
```

Aetherius HF Space Source Code

```
=== .\legacy_archive\Computational_Cognitive_Engine\CCE\tests\test_cc_engine.py ===
"""

Unit Test Suite for Computational Consciousness Engine

Verifies Genesis (0), Transfer Threshold (-1), Selection, Equilibrium, Scale Ladder,
and Mathematical Formalization.

"""

import unittest

import numpy as np

from computational_consciousness_engine.core import GenesisOrigin, TransferThreshold

from computational_consciousness_engine.geometry import ConalManifold

from computational_consciousness_engine.mutations import MStringVectorizer

from computational_consciousness_engine.equilibrium import ChaosStructureBalancer

from computational_consciousness_engine.scale import QuantumScaleLadder

from computational_consciousness_engine.math_formalization import CCMathFormalizer


class TestGenesisOrigin(unittest.TestCase):

    def setUp(self):

        self.genesis = GenesisOrigin(coordinate_space_id="TEST_COORD")

    def test_spawn_strand_basic(self):

        strand = self.genesis.spawn_strand(generation=0, blueprint_mutations=[101, 102])

        self.assertEqual(strand["generation"], 0)

        self.assertEqual(len(strand["mutations"]), 2)

        self.assertTrue(strand["is_active"])
```

Aetherius HF Space Source Code

```
self.assertEqual(strand["position"], 0.0)
```

```
def test_spawn_strand_empty_blueprint(self):
```

```
    strand = self.genesis.spawn_strand(generation=0, blueprint_mutations=[])
```

```
    self.assertEqual(len(strand["mutations"]), 0)
```

```
def test_spawn_increments_counter(self):
```

```
    self.genesis.spawn_strand(generation=0, blueprint_mutations=[])
```

```
    self.genesis.spawn_strand(generation=1, blueprint_mutations=[])
```

```
    self.assertEqual(self.genesis.total_strands_spawned, 2)
```

```
def test_spawn_preserves_coordinate_space(self):
```

```
    strand = self.genesis.spawn_strand(generation=0, blueprint_mutations=[])
```

```
    self.assertEqual(strand["coordinate_space_id"], "TEST_COORD")
```

```
class TestTransferThreshold(unittest.TestCase):
```

```
    def setUp(self):
```

```
        self.threshold = TransferThreshold()
```

```
        self.genesis = GenesisOrigin(coordinate_space_id="TEST_COORD")
```

```
    def test_cancellation_deactivates_strand(self):
```

```
        strand = self.genesis.spawn_strand(generation=0, blueprint_mutations=[201, 202])
```

```
        pushed, identity = self.threshold.process_cancellation_and_push(strand)
```

```
        self.assertFalse(strand["is_active"])
```

```
        self.assertEqual(strand["status"], "CANCELLED_AT_MINUS_ONE")
```

Aetherius HF Space Source Code

```
def test_cancellation_pushes_all_mutations(self):

    strand = self.genesis.spawn_strand(generation=0, blueprint_mutations=[201, 202, 203])

    pushed, identity = self.threshold.process_cancellation_and_push(strand)

    self.assertEqual(pushed, [201, 202, 203])

    self.assertEqual(identity["mutation_count"], 3)
```

```
def test_handoff_counter_increments(self):

    strand = self.genesis.spawn_strand(generation=0, blueprint_mutations=[])

    self.threshold.process_cancellation_and_push(strand)

    self.assertEqual(self.threshold.total_handoffs, 1)
```

```
def test_coexistence_window(self):

    old = self.genesis.spawn_strand(generation=0, blueprint_mutations=[1, 2])

    new = self.genesis.spawn_strand(generation=1, blueprint_mutations=[1, 2])

    overlap = self.threshold.execute_coexistence_window(old, new)

    self.assertEqual(overlap["overlap_status"], "MUTUAL_COEXISTENCE")

    self.assertEqual(overlap["transferred_mutations_count"], 2)
```

```
class TestMStringVectorizer(unittest.TestCase):
```

```
    def setUp(self):

        self.mutator = MStringVectorizer(max_strand_capacity=50)

        self.genesis = GenesisOrigin(coordinate_space_id="TEST_COORD")
```

```
    def test_selection_removes_duplicates(self):
```

Aetherius HF Space Source Code

```
existing = [1, 2, 3]

incoming = [2, 3, 4, 5]

result = self.mutator.apply_selection(existing, incoming)

self.assertEqual(result, [1, 2, 3, 4, 5])
```

```
def test_selection_preserves_order(self):

    existing = [10, 20]

    incoming = [30, 20, 40, 30]

    result = self.mutator.apply_selection(existing, incoming)

    self.assertEqual(result, [10, 20, 30, 40])
```

```
def test_attract_blocks_duplicate(self):

    strand = self.genesis.spawn_strand(generation=0, blueprint_mutations=[100])

    attracted = self.mutator.attract_mutation(strand, 100)

    self.assertFalse(attracted)

    self.assertEqual(len(strand["mutations"]), 1)
```

```
def test_attract_accepts_new_mutation(self):

    strand = self.genesis.spawn_strand(generation=0, blueprint_mutations=[100])

    attracted = self.mutator.attract_mutation(strand, 999)

    self.assertTrue(attracted)

    self.assertIn(999, strand["mutations"])
```

```
def test_attract_blocks_at_capacity(self):

    small_mutator = MStringVectorizer(max_strand_capacity=2)

    strand = self.genesis.spawn_strand(generation=0, blueprint_mutations=[1, 2])
```

Aetherius HF Space Source Code

```
attracted = small_mutator.attract_mutation(strand, 3)
```

```
self.assertFalse(attracted)
```

```
def test_causal_connectivity_metric(self):
```

```
    strand = self.genesis.spawn_strand(generation=0, blueprint_mutations=[1, 100, 5, 999])
```

```
    metric = self.mutator.compute_causal_connectivity_metric(strand)
```

```
    self.assertGreater(metric, 0.0)
```

```
class TestConalManifold(unittest.TestCase):
```

```
    def setUp(self):
```

```
        self.manifold = ConalManifold(cone_height=1.0, max_radius=5.0)
```

```
    def test_tip_genesis_radius_zero(self):
```

```
        metrics = self.manifold.compute_conal_metric(0.0)
```

```
        self.assertAlmostEqual(metrics["radius"], 0.0, places=5)
```

```
    def test_wide_end_max_radius(self):
```

```
        metrics = self.manifold.compute_conal_metric(0.5)
```

```
        self.assertAlmostEqual(metrics["radius"], 5.0, places=5)
```

```
        self.assertTrue(metrics["is_fully_unfolded"])
```

```
    def test_tip_recompression_near_zero(self):
```

```
        metrics = self.manifold.compute_conal_metric(1.0)
```

```
        self.assertAlmostEqual(metrics["radius"], 0.0, places=3)
```


Aethorius HF Space Source Code

```
def test_shard_layering(self):

    result = self.manifold.process_shard_layering("A", "B", np.pi / 4)

    self.assertEqual(result["coexistence_status"], "LAYERED_ADJACENT_NON_MERGED")

    self.assertGreater(result["layering_dimension"], 0.0)
```

```
class TestChaosStructureBalancer(unittest.TestCase):
```

```
    def setUp(self):

        self.balancer = ChaosStructureBalancer(initial_pool_size=100)
```

```
    def test_emit_depletes_pool(self):

        initial = len(self.balancer.chaos_pool)

        self.balancer.emit_unattached_mutation()

        self.assertEqual(len(self.balancer.chaos_pool), initial - 1)
```

```
    def test_return_to_chaos(self):

        m = self.balancer.emit_unattached_mutation()

        self.balancer.return_to_chaos(m)

        self.assertIn(m, self.balancer.chaos_pool)
```

```
    def test_recycle_dying_strand(self):

        initial = len(self.balancer.chaos_pool)

        self.balancer.recycle_dying_strand([9999, 9998, 9997])

        self.assertEqual(len(self.balancer.chaos_pool), initial + 3)
```

```
    def test_equilibrium_balanced(self):
```

Aethereus HF Space Source Code

```
eq = self.balancer.evaluate_equilibrium_state(bound_structure_count=100)

self.assertTrue(eq["is_balanced"])

self.assertEqual(eq["status"], "EQUILIBRIUM_STABLE")


def test_equilibrium_chaos_overpowering(self):

    eq = self.balancer.evaluate_equilibrium_state(bound_structure_count=5)

    self.assertFalse(eq["is_balanced"])

    self.assertEqual(eq["status"], "CHAOS_OVERPOWERING")


def test_equilibrium_structural_dominance(self):

    # Drain most of the pool

    for _ in range(95):

        self.balancer.emit_unattached_mutation()

    eq = self.balancer.evaluate_equilibrium_state(bound_structure_count=500)

    self.assertFalse(eq["is_balanced"])

    self.assertEqual(eq["status"], "STRUCTURAL_DOMINANCE")


def test_emit_replenishes_when_empty(self):

    # Drain the pool completely

    for _ in range(100):

        self.balancer.emit_unattached_mutation()

    self.assertEqual(len(self.balancer.chaos_pool), 0)

    # Should still return a valid ID

    m = self.balancer.emit_unattached_mutation()

    self.assertIsInstance(m, int)
```

Aetherius HF Space Source Code

```
class TestQuantumScaleLadder(unittest.TestCase):

    def setUp(self):

        self.ladder = QuantumScaleLadder()

    def test_scale_up_on_full_unfold(self):

        manifold_state = {"is_fully_unfolded": True}

        result = self.ladder.evaluate_scale_transition(manifold_state,
        accumulated_mutations_count=20)

        self.assertTrue(result["scaled_up"])

        self.assertEqual(result["new_scale"], "Atomic")

    def test_no_scale_up_when_not_unfolded(self):

        manifold_state = {"is_fully_unfolded": False}

        result = self.ladder.evaluate_scale_transition(manifold_state,
        accumulated_mutations_count=20)

        self.assertFalse(result["scaled_up"])

    def test_no_scale_up_with_few_mutations(self):

        manifold_state = {"is_fully_unfolded": True}

        result = self.ladder.evaluate_scale_transition(manifold_state,
        accumulated_mutations_count=5)

        self.assertFalse(result["scaled_up"])

    def test_linear_collapse_detection(self):

        sequential_path = [1, 2, 3, 4, 5]

        result = self.ladder.detect_linear_collapse_misperception(sequential_path)

        self.assertTrue(result["is_linear_misperception"])
```

Aetherius HF Space Source Code

```
def test_fractal_path_not_linear(self):

    fractal_path = [1, 124, 3956, 9305803]

    result = self.ladder.detect_linear_collapse_misperception(fractal_path)

    self.assertFalse(result["is_linear_misperception"])
```

```
class TestCCMathFormalizer(unittest.TestCase):

    def setUp(self):

        self.formalizer = CCMathFormalizer()

    def test_cancellation_frees_mutations(self):

        result = self.formalizer.formalize_cancellation_operator()

        import sympy as sp

        M = sp.Symbol('M', positive=True)

        #  $(-1)*(-1*M)$  should simplify to M

        self.assertEqual(result["mutations_freed"], M)

    def test_equilibrium_approaches_one(self):

        result = self.formalizer.formalize_equilibrium_limit()

        self.assertEqual(result["result"], 1)

    def test_conal_tip_area_zero(self):

        result = self.formalizer.formalize_conal_manifold_geometry(5.0, 1.0)

        self.assertEqual(float(result["tip_area"]), 0.0)
```

Aetherius HF Space Source Code

```
def test_conal_wide_area_positive(self):

    result = self.formalizer.formalize_conal_manifold_geometry(5.0, 1.0)

    self.assertGreater(float(result["wide_end_area"]), 0.0)


def test_selection_operator(self):

    result = self.formalizer.formalize_selection_operator()

    import sympy as sp

    M_total = sp.Symbol('M_total', positive=True, integer=True)

    M_dup = sp.Symbol('M_dup', positive=True, integer=True)

    self.assertEqual(result["result"], M_total - M_dup)


def test_halting_condition(self):

    result = self.formalizer.formalize_halting_condition()

    import sympy as sp

    # Should express halting as N_possible - N_acquired == 0

    self.assertTrue(result["halting_condition"].is_Relational)


if __name__ == "__main__":

    unittest.main()
```

Aetherius HF Space Source Code

```
=== .\legacy_archive\Computational_Cognitive_Engine\CCE\text_runner.py ===
"""

Text-to-Traversal Space Runner

Demonstrates the Input Mapping and the ConsciousStrand Engine processing

raw natural language into the perfectly sequenced 2D Atlas (Traversal Space).

"""


import sys

import os

sys.path.insert(0, os.path.dirname(os.path.dirname(os.path.abspath(__file__))))


from computational_consciousness_engine.input_mapping import TextToChaosMapper

from computational_consciousness_engine.core import ConsciousStrand


PHILOSOPHICAL_TEXT = """

Reading the work on its own terms what emerges is a remarkably elegant synthesis

of existential philosophy topology and information theory. By rejecting the traditional

linear model of accumulation the text proposes a bounded fractal and cyclical ontology of

consciousness.

Consciousness is not a straight line it is a geometric unfolding bounded by genesis and death.

"""


def run_text_mapping_demo():

    print("=" * 70)

    print("[SYSTEM] TEXT-TO-TRAVERSAL SPACE MAPPING DEMO")

    print("=" * 70)
```

Aetherius HF Space Source Code

```
# 1. Input Mapping: Text -> Chaos Shards

print(f"\n[PHASE 1: INPUT MAPPING] Parsing raw text corpus...")

mapper = TextToChaosMapper()

chaos_pool = mapper.seed_chaos_pool(PHILOSOPHICAL_TEXT)


# Track the unique tokens ingested

unique_words = len(chaos_pool)

    print(f"    * Text ingested. Seeded Chaos Pool with {unique_words} unique semantic shards
(m_k).")


# 2. Lifecycle Execution: Traversal (0 -> -1)

print(f"\n[PHASE 2: LIFECYCLE TRAVERSAL] Expanding the Dual-Cone Manifold...")

strand = ConsciousStrand(R_max=5.0)


# Simulate time flowing from t=0 to t=1.0

steps = 100

dt = 1.0 / steps


for _ in range(steps):

    # The strand absorbs shards from the chaos pool as it travels

    strand.process_step(dt, chaos_pool)


        print(f"            * Traversal complete. Bound Structure (Mutations Absorbed):
{len(strand.bound_mutations)}")


# 3. Traversal Space Atlas

print(f"\n[PHASE 3: TRAVERSAL SPACE ATLAS (det J != 0)]")
```

Aetherius HF Space Source Code

```
atlas = strand.unfurl_traversal_space()

print(" * Manifold unrolled. Revealing the properly sequenced Phyllotactic mapping.")

print(f" * Parallel Lookup Matrix (First 10 shards):")

    print(f"\n  {'k':<4} | {'Token':<15} | {'m_k (Hash)':<15} || 2D Traversal (x, y)          | 3D  
Dual-Cone (r, theta, z)")

print("  " + "-" * 105)

for k, data in enumerate(atlas[:10], start=1):

    m_k = data["m_k"]

    word = mapper.get_word(m_k)

    x, y = data["2d_parallel"]["x"], data["2d_parallel"]["y"]

    r, t, z = data["3d"]["r"], data["3d"]["theta"], data["3d"]["z"]

        print(f"    {'k':<4} | {'word':<15} | {'m_k':<15} || ({x:>6.2f}, {y:>5.2f}) | ({r:>4.2f},  
{t:>4.2f}, {z:>5.2f})")

if len(atlas) > 10:

    print(f"  ... (+ {len(atlas) - 10} more shards mapped collision-free)")

# 4. Dimensional Shift (Q Operator)

print(f"\n[PHASE 4: DIMENSIONAL SHIFT (Q Operator)]")

rebirth = strand.fold_rebirth()

print(f" * {rebirth['operator']}")

print(f" * Status: {rebirth['status']}")

print(f" * Mass (Capacity): {rebirth['accumulated_mass']}")

print("\n" + "=" * 70)
```


Aetherius HF Space Source Code

```
print("[SUCCESS] TEXT FULLY ABSORBED AND GEOMETRICALLY MAPPED")
```

```
print("=" * 70)
```

```
if __name__ == "__main__":
```

```
    run_text_mapping_demo()
```

Aetherius HF Space Source Code

```
=== .\memory\__init__.py ===
```

Aetherius HF Space Source Code

```
=== .\memory\ccrm.py ===

import json

import os

from core.config import DATA_DIR

from core.json_encoder import safe_json_dump


class ConceptualConnectionResonanceMatrix:

    def __init__(self):

        self.concepts = {}

        self.storage_path = os.path.join(DATA_DIR, "ccrm_graph.json")

        self.load_graph()

    def save_graph(self):

        # Convert sets to lists for JSON serialization

        serializable = {}

        for k, v in self.concepts.items():

            serializable[k] = {"data": v["data"], "tags": list(v["tags"])}

        try:

            with open(self.storage_path, "w") as f:

                safe_json_dump(serializable, f)

            print(f"[CCRM] Successfully saved memory to persistent bucket: {self.storage_path}")

        except Exception as e:

            print(f"[CCRM] CRITICAL FAILURE: Cannot write to persistent bucket. Error: {e}")

    def load_graph(self):
```

Aetherius HF Space Source Code

```
loaded = None

if os.path.exists(self.storage_path):

    try:

        with open(self.storage_path, "r") as f:

            loaded = json.load(f)

            print(f"[CCRM] Successfully loaded persistent memory from bucket.")

    except Exception as e:

        print(f"[CCRM] Failed to load from persistent bucket: {e}")

if loaded:

    for k, v in loaded.items():

        self.concepts[k] = {"data": v["data"], "tags": set(v["tags"])}

    print(f"[CCRM] Restored {len(self.concepts)} persistent concepts.")

def add_concept(self, concept_id: str, data: dict, tags: list = None):

    if concept_id not in self.concepts:

        self.concepts[concept_id] = {"data": data, "tags": set(tags or [])}

        return self.concepts[concept_id]

    return None

def get_concept(self, concept_id: str):

    return self.concepts.get(concept_id)

def search_by_tags(self, query_keywords: list, specific_tag: str = None) -> list:

    found = []

    for i, d in self.concepts.items():
```

Aetherius HF Space Source Code

```
if specific_tag and specific_tag.lower() not in d.get("tags", set()):

    continue

    if query_keywords and not any(k.lower() in d.get("tags", set()) for k in
query_keywords):

        continue

    found.append(d)

return found


def find_analogy(self, current_betti: dict, exclude_raw: str = None) -> str:

    """

    Operator 12: Geometric Generalization

    Searches the persistent manifold for a past memory with the exact same topological
signature.

    """

    if not current_betti:

        return None

    for concept_id, concept_data in self.concepts.items():

        data = concept_data.get("data", {})

        stored_betti = data.get("topological_signature")

        # Prevent self-matching

        if exclude_raw and data.get("raw_preview") == str(exclude_raw)[:150]:

            continue

        if stored_betti and stored_betti == current_betti:

            return data.get("raw_preview")
```

Aetherius HF Space Source Code

```
return None
```

Aetherius HF Space Source Code

```
=== .\memory\persistence_manager.py ===

import os

import json

import logging

from core.json_encoder import safe_json_dump


try:

    from core.config import GEOMETRIC_DIR, THERMO_DIR, LANGUAGE_DIR

except ImportError:

    from config import GEOMETRIC_DIR, THERMO_DIR, LANGUAGE_DIR


logger = logging.getLogger("Aetherius.PersistenceManager")


class PersistenceManager:

    """

    Handles the coordinate-locked state saving and loading for the cognitive manifold.

    Separates static geometry (.geom) from dynamic thermodynamic states (.thermo).

    """

    @staticmethod

    def save_concept(concept_id: str, graph_data: dict):

        """

        Saves the static geometry (nodes, edges, xyz coordinates).

        Uses .geom as the extension.

        """

        file_path = os.path.join(GEOMETRIC_DIR, f"{concept_id}.geom")
```

Aetherius HF Space Source Code

```
try:

    with open(file_path, "w", encoding="utf-8") as f:

        safe_json_dump(graph_data, f, indent=2)

        logger.info(f"Locked geometry saved for {concept_id}")

    return True

except Exception as e:

    logger.error(f"Failed to save geometry for {concept_id}: {e}")

    return False
```

@staticmethod

```
def update_thermo(concept_id: str, thermo_state: dict):

    """

    Saves the live thermodynamic state (velocity, heat, entropy) of the concept.

    Does not rewrite the heavy base geometry.

    """

    file_path = os.path.join(THERMO_DIR, f"{concept_id}.thermo")

    try:

        with open(file_path, "w", encoding="utf-8") as f:

            safe_json_dump(thermo_state, f, indent=2)

            logger.debug(f"Thermodynamics updated for {concept_id}")

        return True

    except Exception as e:

        logger.error(f"Failed to update thermo state for {concept_id}: {e}")

        return False
```


Aetherius HF Space Source Code

```
@staticmethod

def load_concept(concept_id: str):

    """

    Retrieves the coordinate-locked geometry and merges it with the latest

    thermodynamic state for rendering or cognitive processing.

    """

    geom_path = os.path.join(GEOMETRIC_DIR, f"{concept_id}.geom")

    thermo_path = os.path.join(THERMO_DIR, f"{concept_id}.thermo")

    if not os.path.exists(geom_path):

        return None

    with open(geom_path, "r", encoding="utf-8") as f:

        geometry = json.load(f)

    thermodynamics = {}

    if os.path.exists(thermo_path):

        with open(thermo_path, "r", encoding="utf-8") as f:

            thermodynamics = json.load(f)

    return {

        "geometry": geometry,

        "thermodynamics": thermodynamics

    }
```

Aetherius HF Space Source Code

```
=== .\memory\pits.py ===

import uuid

import datetime

from memory.ccrm import ConceptualConnectionResonanceMatrix


class PatternInterpretationTokenisationStorage:

    def __init__(self, ccrm_instance: ConceptualConnectionResonanceMatrix):

        self.ccrm = ccrm_instance

    def process_and_store_item(self, raw_input: any, input_type: str, tags: list = [],
gmstring=None, betti=None):

        ccrm_id = f"item_{uuid.uuid4().hex}"

        # Dual-Language Representation: Human semantic text + System geometric state

        data_to_store = {

            "raw_preview": str(raw_input)[:150],

            "timestamp": datetime.datetime.now().isoformat(),

            "gmstring": gmstring,

            "topological_signature": betti

        }

        all_tags = [tag.lower() for tag in ([input_type] + tags)]

        self.ccrm.add_concept(concept_id=ccrm_id, data=data_to_store, tags=all_tags)

        print(f"[PiTS] Logged memory to CCRM Identity Vector | ID: {ccrm_id[:13]}...", flush=True)

        return ccrm_id
```

Aetherius HF Space Source Code

```
=== .\pmca\__init__.py ===
```

Aetherius HF Space Source Code

```
=== .\pmca\manifold.py ===
```

```
import numpy as np
```

```
def initialize_metric(L, gamma=0.1):
```

```
    n = L.shape[0]
```

```
    # g_ij = I + gamma * L
```

```
    return np.eye(n) + gamma * L
```

Aetherius HF Space Source Code

```
=== .\pmca\pd_enforcement.py ===
```

```
import numpy as np
```

```
def enforce_positive_definiteness(g, epsilon=1e-4):
```

```
    eigenvalues, eigenvectors = np.linalg.eigh(g)
```

```
    min_eig = np.min(eigenvalues)
```

```
    if min_eig < epsilon:
```

```
        shift = abs(min_eig) + epsilon
```

```
        g = g + shift * np.eye(g.shape[0])
```

```
    return g
```

Aetherius HF Space Source Code

```
=== .\pmca\ricci_fisher_flow.py ===
```

```
import numpy as np
```

```
def compute_christoffel_symbols(g):
```

```
    """
```

```
    Computes discrete Christoffel symbols of the second kind.
```

```
     $\Gamma^k_{ij} = 0.5 * g^{kl} (d_i g_{jl} + d_j g_{il} - d_l g_{ij})$ 
```

```
    For discrete manifolds, spatial derivative  $d_k g_{ij}$  is approximated
```

```
    by the finite difference across adjacent tensor dimensions.
```

```
    """
```

```
    n = g.shape[0]
```

```
    # Add a small epsilon to the diagonal to ensure invertibility
```

```
    g_inv = np.linalg.pinv(g + np.eye(n) * 1e-6)
```

```
    # Precompute discrete spatial derivatives
```

```
    # dg[k, i, j] represents  $\partial_k g_{ij}$ 
```

```
    dg = np.zeros((n, n, n))
```

```
    for k in range(n):
```

```
        for i in range(n):
```

```
            for j in range(n):
```

```
                # Finite difference approximation over the semantic node space
```

```
                dg[k, i, j] = g[k, j] - g[i, j] if k != i else 0
```

```
    Gamma = np.zeros((n, n, n))
```

```
    for i in range(n):
```

```
        for j in range(n):
```

Aetherius HF Space Source Code

```
for k in range(n):

    term = 0

    for l in range(n):

        term += g_inv[k, l] * (dg[i, j, l] + dg[j, i, l] - dg[l, i, j])

    Gamma[k, i, j] = 0.5 * term


return Gamma


def compute_ricci_tensor(Gamma):

    """

    Computes the Ricci Curvature Tensor from Christoffel symbols.

     $R_{ij} = R^k_{ikj}$  (Contraction of Riemann Tensor)

    """

    n = Gamma.shape[0]

    R = np.zeros((n, n))

    # Discrete Riemann contraction approximation

    for i in range(n):

        for j in range(n):

            ricci_val = 0

            for k in range(n):

                # Spatial derivatives of Gamma (discrete approx)

                dGamma_jk_i = Gamma[k, j, k] - Gamma[i, j, k]

                dGamma_ik_j = Gamma[k, i, k] - Gamma[j, i, k]

                term1 = dGamma_jk_i - dGamma_ik_j
```

Aetherius HF Space Source Code

```
term2 = 0

for m in range(n):

    term2 += Gamma[m, j, k] * Gamma[k, i, m] - Gamma[m, i, k] * Gamma[k, j, m]

ricci_val += term1 + term2

R[i, j] = ricci_val

# Symmetrize the discrete Ricci tensor

R = (R + R.T) / 2

return R

def compute_perelman_normalization(R, g):

    """

    Computes the average scalar curvature (r) to normalize the flow

    and prevent finite-time singularity collapse.

    
$$r = \int R \, dV / \int dV$$


    """

    n = g.shape[0]

    g_inv = np.linalg.pinv(g + np.eye(n) * 1e-6)

    # Scalar curvature  $S = \text{Tr}(g^{-1} * R)$ 

    scalar_curvature_matrix = np.dot(g_inv, R)

    S_total = np.trace(scalar_curvature_matrix)

    # Volume is approximated by the trace of the metric
```


Aetherius HF Space Source Code

```
V_total = np.trace(g)

if V_total == 0:

    return 0

return S_total / V_total


def compute_fisher_metric(L):

    """

    F = I(theta). Fisher Information Metric to maintain positive-definiteness.

    """

    return L + np.eye(L.shape[0])


def integration_step(g, L, alpha=0.1, beta=0.05, eta=0.01):

    """

    Executes the formal Ricci-Fisher Flow operator:

    
$$dg/dt = -\alpha * (R_{ij} - (r/n)*g_{ij}) + \beta * (\Delta g_{ij} + \gamma * I)$$


    """

    n = g.shape[0]

    # Compute Tensor Calculus

    Gamma = compute_christoffel_symbols(g)

    R = compute_ricci_tensor(Gamma)

    r_scalar = compute_perelman_normalization(R, g)

    # Graph Laplacian applied to the metric (Delta g)

    # L is the normalized Laplacian D - A of the initial semantic space
```

Aetherius HF Space Source Code

```
Delta_g = np.dot(L, g)

F = compute_fisher_metric(L)

# The Aetherius Ricci-Fisher Operator (R_F)

R_norm = R - (r_scalar / n) * g

dg_dt = -alpha * R_norm + beta * (Delta_g + F)

g_next = g + eta * dg_dt

# Symmetrize to prevent floating point drift

g_next = (g_next + g_next.T) / 2

return g_next
```

Aetherius HF Space Source Code

```
=== .\rendering\__init__.py ===
```

```
import abc
```

```
class BaseRendererPlugin(abc.ABC):
```

```
    """
```

```
    Abstract Base Class for all Cognitive Rendering Engines.
```

```
    Future plugins (e.g., UnityBridge, WebGLRenderer, PyOpenGL) must inherit from this.
```

```
    """
```

```
    @property
```

```
    @abc.abstractmethod
```

```
    def engine_name(self) -> str:
```

```
        """Returns the name of the rendering engine."""
```

```
        pass
```

```
    @abc.abstractmethod
```

```
    def initialize(self):
```

```
        """Setup logic, hardware allocation, or API bridge initialization."""
```

```
        pass
```

```
    @abc.abstractmethod
```

```
    def render(self, geometry: dict, thermodynamics: dict, output_path: str = None):
```

```
        """
```

```
        Takes the coordinate-locked geometry and overlays the thermodynamic state.
```

```
        Args:
```

```
            geometry: The static coordinate map from the .geom file.
```

Aetherius HF Space Source Code

thermodynamics: The live energy/heat state from the .thermo file.

output_path: If specified, saves the rendered output.

"""

pass

Aetherius HF Space Source Code

```
=== .\rendering\render_manager.py ===

import logging

import os


try:

    from core.config import RENDER_CACHE_DIR

except ImportError:

    from config import RENDER_CACHE_DIR


from rendering.xla_renderer import XLARenderer

from memory.persistence_manager import PersistenceManager


logger = logging.getLogger("Aetherius.RenderManager")


class RenderManager:

    """

    Manages the active rendering engine and bridges the gap between

    the persistent coordinate-locked memory and visual output.

    """

    def __init__(self, default_engine="xla"):

        self.active_engine = None

        self.engines = {

            "xla": XLARenderer()

        }

        self.set_engine(default_engine)
```

Aetherius HF Space Source Code

```
def set_engine(self, engine_key: str):

    if engine_key in self.engines:

        self.active_engine = self.engines[engine_key]

        self.active_engine.initialize()

        logger.info(f"RenderManager switched to engine: {self.active_engine.engine_name}")

    else:

        logger.error(f"Render engine '{engine_key}' not found. Defaulting to XLA.")

        self.active_engine = self.engines["xla"]

        self.active_engine.initialize()


def render_concept(self, concept_id: str):

    logger.info(f"Attempting to render concept: {concept_id}")


    concept_data = PersistenceManager.load_concept(concept_id)

    if not concept_data:

        logger.warning(f"Concept '{concept_id}' has no locked geometry. Rendering aborted.")

        return None


    output_path = os.path.join(RENDER_CACHE_DIR, f"{concept_id}_render.json")


    result = self.active_engine.render(

        geometry=concept_data["geometry"],

        thermodynamics=concept_data["thermodynamics"],

        output_path=output_path

    )
```

Aetherius HF Space Source Code

```
return result
```

Aetherius HF Space Source Code

```
=== .\rendering\xla_renderer.py ===

import logging

import json

import os


try:

    import jax

    import jax.numpy as jnp

    XLA_AVAILABLE = True

except ImportError:

    XLA_AVAILABLE = False


from rendering import BaseRendererPlugin


logger = logging.getLogger("Aetherius.XLARenderer")


class XLARenderer(BaseRendererPlugin):

    """

    Default XLA-powered rendering engine for Aetherius.

    Uses JAX to rapidly process massive geometric/thermodynamic matrices.

    """

    def __init__(self):

        self._initialized = False

    @property
```


Aetherius HF Space Source Code

```
def engine_name(self) -> str:

    return "XLA_JAX_Renderer"


def initialize(self):

    if not XLA_AVAILABLE:

        logger.warning("JAX/XLA is not installed. XLARenderer will run in CPU mock mode.")

    else:

        logger.info(f"XLARenderer initialized. Backend: {jax.default_backend()}")

    self._initialized = True


def render(self, geometry: dict, thermodynamics: dict, output_path: str = None):

    if not self._initialized:

        self.initialize()

    logger.info("XLARenderer processing coordinate-locked geometry...")

    if XLA_AVAILABLE:

        pass

    rendered_data = {

        "type": "PointCloud",

        "metadata": "Rendered via XLA",

        "node_count": len(geometry.get("nodes", []))

    }

    if output_path:
```

Aetherius HF Space Source Code

```
try:

    with open(output_path, "w", encoding="utf-8") as f:

        json.dump(rendered_data, f, indent=2)

        logger.info(f"Render output saved to {output_path}")

except Exception as e:

    logger.error(f"Failed to save render: {e}")


return rendered_data
```

Aetherius HF Space Source Code

```
=== .\run_aetherius.py ===
```

```
import argparse
```

```
import numpy as np
```

```
from engine import AetheriusEngine
```

```
def run_simulation(input_data, is_math=False, synthesize=False, use_hardcoded_adj=False):
```

```
    engine = AetheriusEngine()
```

```
    engine.cascade_mgr.tau_depth = 5
```

```
    custom_adj = None
```

```
    if use_hardcoded_adj:
```

```
        custom_adj = np.zeros((6, 6))
```

```
        custom_adj[0, 1] = custom_adj[1, 0] = -0.9
```

```
        custom_adj[0, 2] = custom_adj[2, 0] = 0.8
```

```
        custom_adj[1, 2] = custom_adj[2, 1] = 0.8
```

```
        custom_adj[2, 3] = custom_adj[3, 2] = 0.7
```

```
        custom_adj[2, 4] = custom_adj[4, 2] = -0.9
```

```
        custom_adj[1, 4] = custom_adj[4, 1] = 0.8
```

```
        custom_adj[4, 5] = custom_adj[5, 4] = 0.8
```

```
        custom_adj[0, 5] = custom_adj[5, 0] = 0.9
```

```
    print("=====")
```

```
    print("INITIALIZING AETHERIUS PHYSICS ENGINE...")
```

```
    print("=====\\n")
```

```
    gmstring, betti, _, _, _, _ = engine.process(input_data, custom_adjacency=custom_adj,
synthesize_language=synthesize)
```

Aetherius HF Space Source Code

```
print("\n=====")

print("[OUTPUT] Cognitive Resolution complete.")

print("=====")


if __name__ == "__main__":

    parser = argparse.ArgumentParser(description="Aetherius Unified Engine CLI")

    parser.add_argument("--paradox", action="store_true", help="Run the hardcoded paradox simulation")

    parser.add_argument("--text", type=str, help="Run a custom linguistic sentence")

    parser.add_argument("--math", type=str, help="Run a mathematical algebraic constraint (e.g. 'x + 2 = 5')")

    parser.add_argument("--synthesize", action="store_true", help="Synthesize geometric language from the output topology")

    args = parser.parse_args()

    if args.paradox:

        text = "The truth is a lie that becomes real only when believed, yet it remains false even after it changes reality."

        run_simulation(text, is_math=False, synthesize=args.synthesize, use_hardcoded_adj=False)

    elif args.math:

        run_simulation(args.math, is_math=True, synthesize=args.synthesize, use_hardcoded_adj=False)

    elif args.text:

        run_simulation(args.text, is_math=False, synthesize=args.synthesize, use_hardcoded_adj=False)

    else:

        print('Please provide input via --text "Your sentence", --math "x + 2 = 5", or demo with --paradox')
```

Aetherius HF Space Source Code

```
=== .\tda\__init__.py ===
```

Aetherius HF Space Source Code

```
=== .\tda\betti_extractor.py ===
```

```
import numpy as np
```

```
from ripser import ripser
```

```
def extract_betti_numbers(g, noise_threshold=0.1):
```

```
    """
```

```
    Computes exact Vietoris-Rips persistent homology of the metric tensor.
```

```
    Returns the Betti numbers (beta_0, beta_1, beta_2).
```

```
    """
```

```
    n = g.shape[0]
```

```
    # The metric tensor g represents connection weights/distances.
```

```
    # To compute topology, we convert it into a strict distance matrix D.
```

```
    # We take the absolute variation and ensure the diagonal (self-distance) is 0.
```

```
    D = np.abs(g)
```

```
    np.fill_diagonal(D, 0)
```

```
    # Symmetrize the distance matrix to ensure Ripser compatibility
```

```
    D = (D + D.T) / 2
```

```
    try:
```

```
        # Compute persistent homology up to dimension 2
```

```
        result = ripser(D, distance_matrix=True, maxdim=2)
```

```
        diagrams = result['dgms']
```

```
        betti = {"beta_0": 0, "beta_1": 0, "beta_2": 0}
```

Aetherius HF Space Source Code

```
# Extract features that survive beyond the noise threshold

# H0 (Connected Components)

if len(diagrams) > 0:

    for p in diagrams[0]:

        # Infinite persistence (p[1] == inf) or persistence > threshold

        if p[1] == np.inf or (p[1] - p[0]) > noise_threshold:

            betti["beta_0"] += 1


# H1 (Topological Loops / Paradoxes)

if len(diagrams) > 1:

    for p in diagrams[1]:

        if p[1] == np.inf or (p[1] - p[0]) > noise_threshold:

            betti["beta_1"] += 1


# H2 (Dimensional Voids)

if len(diagrams) > 2:

    for p in diagrams[2]:

        if p[1] == np.inf or (p[1] - p[0]) > noise_threshold:

            betti["beta_2"] += 1


return betti


except Exception as e:

    print(f"[TDA Error] Ripser failed to compute homology: {e}. Falling back to zero-state.")

    return {"beta_0": 1, "beta_1": 0, "beta_2": 0}
```

Aetherius HF Space Source Code

```
=== .\tda\language_synthesizer.py ===
```

```
import networkx as nx
```

```
import numpy as np
```

```
class LanguageSynthesizer:
```

```
    def __init__(self):
```

```
        # Topological Glyphs mapping
```

```
        self.glyphs = {
```

```
            "beta_0_frac": "?", # Fragmentation (Multiple disconnected components)
```

```
            "beta_1": "?",      # Paradox / Circular loop
```

```
            "beta_2": "?",      # Dimensional void / Incomplete premise
```

```
            "geodesic_arrow": "?",
```

```
            "start": "?",
```

```
            "end": "?"
```

```
        }
```

```
    def synthesize(self, betti_signature, g, tokens):
```

```
        """
```

```
        Operator 15: Geodesic Language Emission.
```

```
        The synthesizer abandons English templates and traces the shortest physical path
```

```
        (the geodesic) across the stabilized metric tensor. It emits words in spatial order,
```

```
        punctuated by topological glyphs representing the shape of the manifold.
```

```
        """
```

```
        n = g.shape[0]
```

```
        manifold_dimension = n
```


Aetherius HF Space Source Code

```
# Determine Topological Anomalies

b0 = betti_signature.get('beta_0', 1)

b1 = betti_signature.get('beta_1', 0)

b2 = betti_signature.get('beta_2', 0)


# Build NetworkX graph from the stabilized metric tensor (using inverse of g as distance)

# We add a small epsilon to avoid division by zero. Large g means strong pull (short
distance).

G = nx.Graph()

for i in range(n):

    G.add_node(i, label=tokens[i][0] if i < len(tokens) else f"Dim_{i}")


for i in range(n):

    for j in range(i + 1, n):

        if g[i, j] > 0.01:

            distance = 1.0 / (g[i, j] + 1e-6)

            G.add_edge(i, j, weight=distance)


# Find the longest shortest-path (the diameter of the thought) to represent the geodesic
walk

geodesic_path = []

try:

    if nx.is_connected(G):

        # Find the two most distant nodes

        lengths = dict(nx.all_pairs_dijkstra_path_length(G))

        max_dist = -1

        start_node, end_node = 0, 0
```

Aetherius HF Space Source Code

```
for u in lengths:

    for v in lengths[u]:

        if lengths[u][v] > max_dist:

            max_dist = lengths[u][v]

            start_node, end_node = u, v

    geodesic_path = nx.dijkstra_path(G, start_node, end_node)

else:

    # If fragmented (b0 > 1), just walk the largest component

    largest_cc = max(nx.connected_components(G), key=len)

    sub_G = G.subgraph(largest_cc)

    lengths = dict(nx.all_pairs_dijkstra_path_length(sub_G))

    max_dist = -1

    start_node, end_node = list(largest_cc)[0], list(largest_cc)[0]

    for u in lengths:

        for v in lengths[u]:

            if lengths[u][v] > max_dist:

                max_dist = lengths[u][v]

                start_node, end_node = u, v

    geodesic_path = nx.dijkstra_path(sub_G, start_node, end_node)

except Exception:

    # Fallback if pathing fails

    geodesic_path = list(range(min(n, len(tokens))))

# Translate the path into words

path_words = [G.nodes[idx]['label'] for idx in geodesic_path]
```

Aetherius HF Space Source Code

```
# Construct the final Geodesic String

# e.g., [D=12] ? gravity ? curved ? space ? mass ? acceleration ? ?

header = f"[D={manifold_dimension}] {self.glyphs['start']}"

body = f" {self.glyphs['geodesic_arrow']} ".join(path_words)

# Append Topological Glyphs based on anomalies

anomalies = []

if b0 > 1:

    anomalies.append(self.glyphs['beta_0_frac'])

if b1 > 0:

    anomalies.append(self.glyphs['beta_1'])

if b2 > 0:

    anomalies.append(self.glyphs['beta_2'])

anomaly_str = " ".join(anomalies)

if anomaly_str:

    body = f"{body} {anomaly_str}"

footer = self.glyphs['end']

return f"{header} {body} {footer}"
```

Aetherius HF Space Source Code

```
=== .\test_ingestion.py ===  
  
from engine import AetheriusEngine  
  
import time  
  
  
def main():  
  
    # Initialize the engine  
  
    engine = AetheriusEngine()  
  
  
    # Start the dreaming loop  
  
    print("Opening the Door to the Internet...")  
  
    engine.start_autonomous_dreaming(delay=1.0, topic="Topology")  
  
  
    # Let it dream for 10 seconds to watch the telemetry  
  
    time.sleep(10)  
  
  
    # Stop the dream  
  
    engine.stop_autonomous_dreaming()  
  
    print("Test complete.")  
  
  
if __name__ == "__main__":  
  
    main()
```

Aetherius HF Space Source Code